

SPACE RACE

DATA SCIENCE REPORT



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Space Race Report (better header)

This project analyzes a comprehensive dataset on global space missions sourced from nextspaceflight.com, covering the period from the start of the space race in 1957 to modern space exploration.

The dataset contains detailed information on mission results, launch frequency, costs, and participating organizations, enabling us to analyze both technological progress and geopolitical influences over time. Using data science tools such as pandas, matplotlib, and Plotly, this report transforms raw launch data into meaningful visual insights.

The analysis focuses on key questions such as which countries and organizations played a leading role during different eras, how mission costs and success rates evolved, and how historical events such as the Cold War shaped space exploration.

Ultimately, this project aims to demonstrate how data visualization and analysis can shed light not only on trends in space exploration, but also on the broader history of human innovation and competition beyond Earth.

Source Data Files (CSV Format)

Mission Launches: github.com

Imported Libraries

```
import pandas as pd
import numpy as np
import plotly.express as px
import matplotlib.pyplot as plt
from pandas.plotting import register_matplotlib_converters
from sklearn.linear_model import LinearRegression
import matplotlib.dates as mdates
import iso3166
import plotly.graph_objects as go
```

Display Formatting for Financial Data

To improve readability and ensure consistent formatting, we are going to configure pandas to display floating-point numbers and values in a more appropriate way.

```
# Create locators for ticks:
register_matplotlib_converters()

# Notebook Presentation:
pd.options.display.float_format = '{:,.2f}'.format
pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)
```

Acquiring the Source and Exploring Initial Dataset

```
#Load the Data
df = pd.read_csv('csv_files/mission_launches.csv')
```

The primary dataset for this analysis comes from Kaggle and consists of launch logs compiled from the internet via the aerospace tracking database Next Spaceflight.

The data was collected in August 2020 and provides a comprehensive historical overview of all space launches recorded from 1957 up to the time of data collection.

While the data is highly reliable for tracking launch frequencies, geographic locations, and mission success rates, **the financial analysis is limited** due to the inherent lack of cost data for classified government missions from the Cold War era.

Obtaining the Structure of the Dataset

We acquire the numbers of rows (entries) and the columns (features) by using `.shape()` function. Besides `.column()` function gives us the names of the features that are used in the dataset to classify each entry.

```
print("\nShape of the DataFrame:")
print(df.shape)

print("\nColumns of the Dataframe:")
print(df.columns)
```

Shape of the DataFrame:

```
(4324, 9)
```

Columns of the DataFrame:

```
Index(['Unnamed: 0.1', 'Unnamed: 0', 'Organisation', 'Location', 'Date',  
      'Detail', 'Rocket_Status', 'Price', 'Mission_Status'],  
      dtype='str')
```

As reflected above, we have 4324 different entries which show the rocket launches by all involved countries between 1957 and 2020. Furthermore, the dataset possesses 9 columns which 7 of them are relevant for the analysis.

Meanings of the Features (Columns) in the DataFrame:

Organization: This represents the specific company or state agency (such as SpaceX, CASC, or Roscosmos) that manufactured and operated the launch vehicle.

Location: This indicates the specific launch pad and geographical site where the rocket blasted off. It usually includes the facility name, region, and the country.

Date: This records the exact day, month, year, and time that the launch occurred.

Detail: This column contains the specific name and model of the rocket used for the mission. It also frequently includes the name of the payload or satellite that the rocket was carrying into orbit.

Rocket_Status: This shows whether the specific model of rocket used for the mission is still actively flying or has been permanently retired as of August 2020.

Price: This represents the estimated cost of the rocket launch recorded in millions of US Dollars (USD). As previously noted, this column contains many missing values for older or classified government missions.

Mission_Status: This is a categorical variable indicating the final outcome of the launch, such as whether it was a complete success, a failure, or a partial failure. This is the metric what we will use to calculate success rates over time.

Removing Irrelevant Columns

We are going to remove unnecessary index-related columns that were automatically created during the data export to keep the dataset clear, focused, and ready for analysis.

```
df = df.drop('Unnamed: 0.1', axis=1)
df = df.drop('Unnamed: 0', axis=1)
```

Initial Data Exploration

The dataset is loaded into a pandas DataFrame, and we are going to display its first few rows with using `.head()` function to provide an initial overview of its structure, variables, and overall content.

```
#Load the Data
df = pd.read_csv('csv_files/space_files/mission_launches.csv')
```

First rows of the DataFrame:

Organisation	Location	Date	Detail	Rocket_Status	Price	Mission_Status
SpaceX	LC-39A, Kennedy Space Center, Flori...	Fri Aug 07, 2020 05:12	Falcon 9 Block 5 Starlink V1 L9 & BlackSky	StatusActive	50.0	Success
CASC	Site 9401 (SLS-2), Jiuquan Satellite Launc...	Thu Aug 06, 2020 04:01	Long March 2D Gaofen-9 04 & Q-SAT	StatusActive	29.75	Success
SpaceX	Pad A, Boca Chica, Te...	Tue Aug 04, 2020 23:57	Starship Prototype 150 Meter Hop	StatusActive	NaN	Success
Roscosmos	Site 200/39, Baikonur Cosmodrome, Ka...	Thu Jul 30, 2020 21:25	Proton-M/Briz-M Ekspress-80 & Ekspress-103	StatusActive	65.0	Success
ULA	SLC-41, Cape Canaveral AFS, Flor...	Thu Jul 30, 2020 11:50	Atlas V 541 Perseverance	StatusActive	145.0	Success

The preview shows that each row represents a single launch, including key attributes such as organization, location, date, mission details, cost, and mission status.

It is noticeable that, although most orders are marked as successful, some price data is missing, suggesting that the cost data is incomplete and may need to be further processed. Furthermore, the dataset already reflects the global nature of modern space exploration, as organizations such as SpaceX, Roscosmos, and CASC appear right from the start.

The values in the “Price” column are listed in millions of U.S. dollars. For example, for the entry “SpaceX - Falcon 9 Block 5” The number 50.0 means that this launch cost 50 million dollars.

Checking for Missing and Duplicated Entries

In this step, we are going to inspect the dataset for missing or duplicate entries that could compromise the reliability of the analysis

```
print("\nIs there any missing values?")
print(df.isna().values.any())

print("\nIs there any duplicated values?")
print(df.duplicated().values.any())
```

```
Is there any missing values?
```

```
True
```

```
Is there any duplicated values?
```

```
True
```

The results show that there are both missing values and duplicates, suggesting that further data cleaning is necessary.

We will take a closer look at these points to ensure that the dataset is accurate and consistent before proceeding with the analysis.

Inspecting Duplicate Records

Using `duplicated()`, we perform a duplicate check to determine whether the dataset contains duplicate rows, thereby ensuring data integrity prior to analysis.

```
print("\nThe Amount of Duplicated Entries:")
print(df.duplicated().values.any().sum())
```

```
The Amount of Duplicated Entries:
```

```
1
```

The results show that there is only one duplicate entry, which can be safely removed without affecting the entire dataset.

```
# There is just one duplicates, simply drop it.
df.drop_duplicates(keep='last', inplace=True)
```

Acquiring the Overview of the DataFrame

We use *info()* function to get the structure of the DataFrame and the overview of the feature(columns) their type and content.

```
print("\nOverview of the DataFrame:")
print(df.info())
```

Overview of the DataFrame:

```
<class 'pandas.DataFrame'>
Index: 4323 entries, 0 to 4323
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Organisation    4323 non-null   str
1   Location        4323 non-null   str
2   Date            4323 non-null   str
3   Detail          4323 non-null   str
4   Rocket_Status   4323 non-null   str
5   Price           963 non-null    str
6   Mission_Status  4323 non-null   str
dtypes: str(7)
```

All columns are stored as strings, and while most fields are fully populated, the “Price” column contains a significant number of missing values; there are only 963 non-zero entries.

This is to be expected, as many space agencies and organizations particularly in the early years of the space race which is called cold war, did not publicly disclose the costs of their missions; this circumstance is carefully taken into account in the analysis.

Date Transformation: Converting Strings to Datetime Format

The “Date” column is originally stored as text with various formatting options, which makes direct time-based analysis difficult.

To resolve this issue, we are going to write and apply a custom parsing function to convert these strings into standardized datetime objects, thereby ensuring the consistency of all entries.

This step is essential for enabling chronological analyses such as annual trends, time series visualizations, and comparisons between different time periods.

```
def parse_date(x):
    try:
        return pd.to_datetime(x, format='%a %b %d, %Y %H:%M UTC', utc=True)
    except:
        try:
            return pd.to_datetime(x, format='%a %b %d, %Y', utc=True)
        except:
            return pd.NaT

# Applying Formatting:
df['Date'] = df['Date'].apply(parse_date)
df['Date'] = df['Date'].dt.tz_localize(None)
```

Price Conversion: Transforming Strings into Numeric Values

We are going to convert the “Price” column from text format to numerical values (floats) by removing formatting characters such as commas and converting invalid entries to missing values.

This enables accurate mathematical calculations, such as determining averages, totals, and distributions of operating costs.

```
df['Price'] = pd.to_numeric(
    df['Price'].str.replace(',', '', regex=False),
    errors='coerce'
)
```

Price Analysis and Descriptive Statistics

In this step, we are going to calculate summary statistics for the “Price” column to capture the overall distribution, central tendency, and dispersion of mission costs.

This allows us to quickly identify typical cost ranges and outliers and determine the extent to which mission costs vary within the dataset.

```
# Descriptive Statistics
print("\nDescriptive Statistics of the DataFrame:")
print(df.Price.describe())
```

Descriptive Statistics of the DataFrame:

```
count      963.00
mean       153.92
std        288.57
min         5.30
25%        40.00
50%        62.00
75%       164.00
max       5,000.00
Name: Price, dtype: float64
```

The results show that the average mission cost is approximately \$154 million, but the very high standard deviation indicates significant variation in spending.

The median value (\$62 million) is significantly lower than the mean, suggesting that a few extremely expensive missions (outliers) are skewing the average upward.

While many missions fall within a moderate range (25% under \$40 million and 75% under \$164 million), the maximum total cost of \$5 billion (5,000 million dollars) also highlights the existence of exceptionally costly space programs.

Feature Engineering with Location Data

The dataset contains a “Location” column in which detailed information about launch sites—such as specific launch pads, space centers, and countries—is consolidated into a single string.

In this step, we are going to extract the country information from these entries and create a new “Country” column to better structure the data and prepare it for analysis.

This transformation enables us to easily compare the launch activities of different countries and perform clearer geographic analyses.

```
df["Country"] = df["Location"].astype(str).str.split(",").str[-1].str.strip()
```

Standardizing Country Information for Analysis

The extracted country data contains inconsistencies and ambiguous place names (such as “seas” or “test sites”) that are unsuitable for meaningful comparisons at the country level.

Below are two examples of these entries from the dataset:

Organisation	Location	Date	Detail
CASC	Tai Rui Barge, Yellow Sea	Wed Jun 05, 2019 04:06 UTC	Long March 11H Jilin-1
IRGC	Launch Platform, Shahrud Missile Test Site	Wed Apr 22, 2020 03:59 UTC	Qased Noor

To address this, we assign these entries to the appropriate countries, ensuring that each launch is correctly assigned to a recognizable and consistent country.

We define the following conversions which we need to apply:

```
# Define mappings for ambiguous Locations
location_mapping = {
    "Russia": "Russian Federation",
    "New Mexico": "USA",
    "Yellow Sea": "China",
    "Shahrud Missile Test Site": "Iran",
    "Pacific Missile Range Facility": "USA",
    "Barents Sea": "Russian Federation",
    "Gran Canaria": "USA"
}
```

Applying the Mapping with using `.replace()` function:

```
# Apply Location mapping
df["Country"] = df["Country"].replace(location_mapping)
```

This step improves the reliability of the geographic analysis by preventing fragmented or misleading categories, allowing us to gain clearer insights when comparing market launch activities across different countries.

Adding ISO Country Codes for Geographical Analysis

To enable complex geographic visualizations such as choropleth maps, we supplement the dataset with ISO Alpha-3 country codes for each entry. These standardized codes allow mapping libraries to correctly identify and display countries using their international abbreviations.

Handling Special Cases in ISO Conversion

We use the “iso3166” library to automatically convert country names into the corresponding ISO codes, thereby ensuring consistency throughout the dataset.

However, since some entries do not conform to the standard or cannot be recognized directly (e.g., oceans or unusual naming conventions), we define custom mappings as needed to assign the correct ISO codes.

```
# Plotting "ISO" names to the data:
# Define missing ISO codes
missing_iso_mapping = {
    "Iran": "IRN",
    "North Korea": "PRK",
    "Pacific Ocean": "PAC", # Custom code for non-country regions
    "South Korea": "KOR",
    "USA": "USA"
```

Generating ISO Codes for Each Entry:

A custom function is applied to map each country name to its ISO Alpha-3 code, combining both library-based conversions and manual mappings.

This step ensures that every record in the dataset is correctly formatted for geographic analysis and visualization.

```
# Function to get Alpha-3 country codes
def get_iso_alpha3(country):
    country = country
    if country in missing_iso_mapping:
        return missing_iso_mapping[country] # Return manually assigned code if
available
    try:
        return iso3166.countries_by_name[country.upper()].alpha3
```

```

except KeyError:
    return missing_iso_mapping[country] # Handle missing cases gracefully

# Convert country names to ISO Alpha-3 codes
df["ISO"] = df["Country"].map(get_iso_alpha3)

```

Extracting Space Programs (Rocket Type) from Mission Details

The “Details” column follows a standard format: “Launch vehicle | Payload (mission name)”, where the first part specifies the launch vehicle used and the second part describes the mission or payload it is carrying.

In this step, we extract information about the rockets (Space programs and launch vehicles) and create a new column called “Rocket_Type,” which allows us to analyze the performance, frequency, and cost distribution of various rocket programs. This is particularly useful because, in the dataset, mission costs are often associated with the rocket itself and spread across multiple launches, making an analysis at the rocket level more meaningful than one at the payload level.

From a data science perspective, this transformation is achieved through feature engineering by splitting the string values at the “|” delimiter and selecting the first component, which corresponds to the rocket's name. This structured data extraction enables more detailed grouping, summarization, and comparison of launch vehicles such as the Falcon 9 and the Long March 2D, ultimately improving the depth and clarity of the analysis.

```

# Adding Rocket Types to the DataFrame:
# This creates two new columns by splitting the text at the pipe '|' symbol
df['Rocket_Type'] = df['Detail'].str.split(' | ', regex=False,
expand=True)[0].str.strip()

```

Displaying the DataFrame with appended Features

```

print("\nDataFrame with new Features:")
print(df.head())

```

DataFrame with new Features:

Organisation	Location	Date	Detail	Rocket_Status	Price	Mission_Status	Country	ISO	Rocket_Type
SpaceX	...	2020-08-07 05:12:00	...	...	50.00	...	USA	USA	Falcon 9 Block 5
CASC	...	2020-08-06 04:01:00	...	...	29.75	...	China	CHN	Long March 2D
SpaceX	...	2020-08-04 23:57:00	...	...	NaN	...	USA	USA	Starship Prototype
Roscosmos	...	2020-07-30 21:25:00	...	...	65.00	...	Kazakhstan	KAZ	Proton-M/Briz-M
ULA	...	2020-07-30 11:50:00	...	...	145.00	...	USA	USA	Atlas V 541

We addressed key data quality issues by handling missing and duplicate values and converting important fields—such as dates and prices—into formats suitable for analysis. In addition, we have standardized inconsistent entries and made the dataset more structured and easier to compare by introducing new attributes, including country names, ISO codes, and missile types.

Now that these preprocessing steps have been completed, the dataset has been cleaned and enriched and is ready for meaningful analysis and visualization.

Analysis of Leading Space Organizations

In this section, we are going to analyze the top 20 organizations with the most launches in the history to identify the dominant players in the global space industry and understand how launch activity has evolved over time. Examining these organizations provides valuable insights into the evolution of space exploration—from the government-dominated years of the space race to the rise of modern private companies such as SpaceX.

Ultimately, this section is intended to use the raw data to provide a clearer picture of the organizations that have shaped the history of modern spaceflight.

Grouping the DataFrame by Organizations and Counting Launch Activity

To calculate the number of launches performed by each organization, the dataset is grouped by both organization name and country using the pandas `groupby()` function.

The grouped entries are then counted using the `.size()` method, converted into a structured DataFrame using `.reset_index()`, and finally sorted in descending order to identify the 20 organizations with the highest launch activity.

```
# Number of Launches per Company
df_by_organisation = (
    df.groupby(['Organisation', 'Country'])
      .size()
      .reset_index(name='Total_Launches')
)

df_first_20 = df_by_organisation
                .sort_values('Total_Launches', ascending=False).head(20)
```

Creating the Launch Distribution Bar Chart

The Plotly's `px.bar()` is used to create a horizontal bar chart that visually compares the total number of launches by different organizations. This visualization improves readability for long organization names while using color coding and interactive metadata to make comparisons clearer and more meaningful.

code:

```
h_bar = px.bar(
    df_first_20,
    x='Total_Launches',
    y='Organisation',
    orientation='h',
    text='Total_Launches', color='Organisation',
    color_discrete_sequence=px.colors.qualitative.G10,
    custom_data=['Country'],
    labels={'Organisation': 'Organisation', 'Total_Launches': 'Total Launches'})

# Designing the Layout
h_bar.update_layout(
    plot_bgcolor='white', paper_bgcolor='white',
    font_family='Arial, Helvetica, sans-serif',
    title=dict(text="Top Space Organizations by Launches",
               font=dict(family='Arial, Helvetica, sans-serif', size=28,
                        color='#D14D72'), x=0.03),

    xaxis_title='', yaxis_title='',
    yaxis=dict(
        categoryorder='total ascending',
```

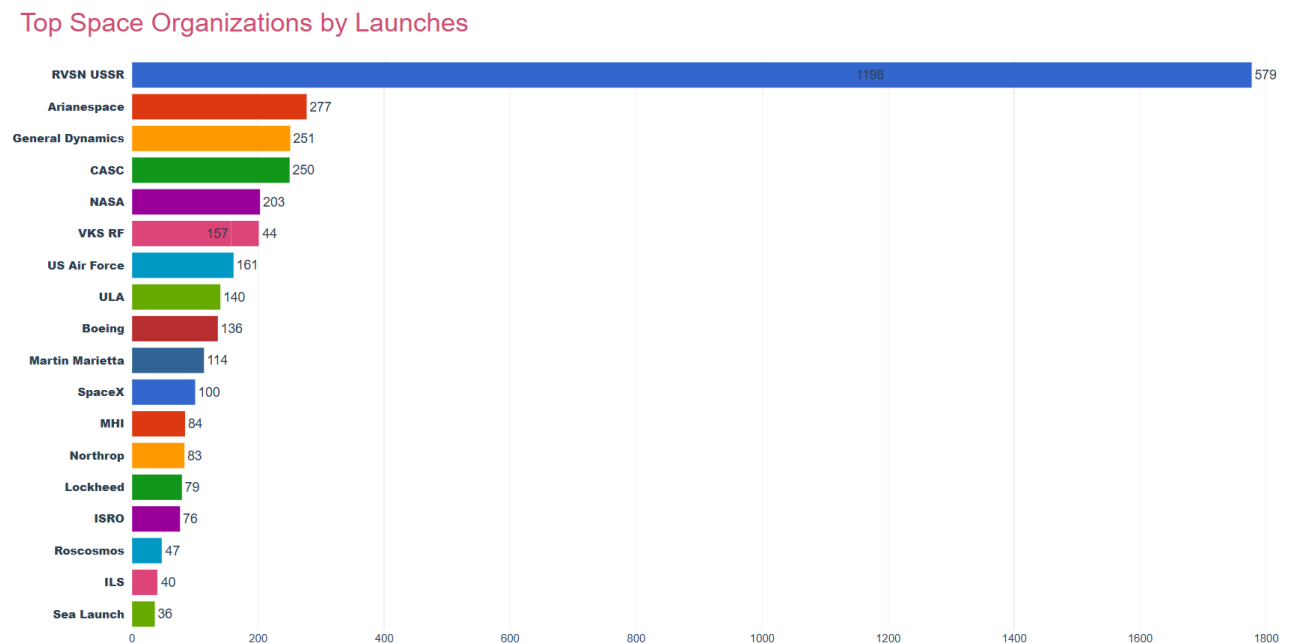
```

        showgrid=False, showline=False,
        ticks='outside', ticklen=5, tickcolor='white',
        tickfont=dict(family='Arial Black', size=12, color='#2c3e50')
    ),
    xaxis=dict(showgrid=True, gridcolor='#f0f0f0', showticklabels=True),
    coloraxis_showscale=False, showlegend=False, height=750
)

# Alignments for the Labels and bars
h_bar.update_traces(
    textposition='outside', textfont=dict(size=14, color='#2c3e50'),
    marker_line_width=0, cliponaxis=False,
    hovertemplate=(
        '<b style="font-size: 14px;">{%y}</b><br>' +
        '<span style="font-size: 13px;">Total Launches: </span>' +
        '<b style="font-size: 14px;">{%x}</b>'<br>' +
        '<span style="font-size: 13px;">Country: </span>' +
        '<b style="font-size: 14px;">{%customdata[0]}</b>' +
        '<extra></extra>'
    )
)
h_bar.show()

```

Graph:



Interactive Graph Link: [Top Space Organizations by Launches](#)

The graph illustrates the overwhelming dominance of the Soviet RVSN organization, whose launch figures far exceed those of all other organizations. This reflects the enormous technological and political pressure that prevailed at the start of the space race, when a high launch frequency often took precedence over economic efficiency or environmental sustainability.

Another notable observation is the strong presence of both traditional government agencies, such as NASA, and modern commercial players, such as SpaceX, which illustrates how the space industry has gradually evolved from a government-controlled geopolitical competition into a commercially driven technology sector.

The relatively low number of launches by new and private companies, despite SpaceX's significant influence, also underscores an important shift in the industry: Today, these companies are increasingly focusing on reusable rockets, payload capacity, and mission efficiency, rather than simply maximizing the number of launches.

Analysis of Mission Success Rates

The success rate of space missions is one of the most important indicators of technological maturity, technical reliability, and operational experience in the aerospace industry. In this section, we are going to analyze the distribution of successful and failed launches to better understand how reliable space missions have become over the course of modern space exploration.

Analyzing mission results not only helps determine the overall performance of launch systems but also provides insights into how decades of technological progress, testing, and international competition have contributed to improving launch safety and mission performance.

Calculating Mission Outcome Distribution

The dataset is grouped by mission status using the pandas `.groupby()` function, and the number of entries in each category is counted to determine how frequently different mission outcomes occurred.

```
df_by_mission_status = (df.groupby('Mission_Status', as_index=False)
                          .size().rename(columns={'size': 'Count'}))
```


Creating a Pie Chart for Mission Outcomes

Using Plotly, a pie chart is created to visually represent the proportion of successful, failed, and partially failed missions in the dataset. Custom color coding and interactive tooltips improve readability and make it easier to interpret the distribution of mission outcomes.

code:

```
color_map = {'Success': '#00B8A9', 'Failure': '#F6416C', 'Partial Failure': '#FFDE7D', 'Prelaunch Failure': '#E4C59E'}
```

```
colors = [color_map.get(status, '#CCCCCC') for status in df_by_mission_status['Mission_Status']]
```

```
# This dynamically pulls the 'Failure' slice out by 10% (0.1)
pull_values = [0.1 if status == 'Failure' else 0 for status in df_by_mission_status['Mission_Status']]
```

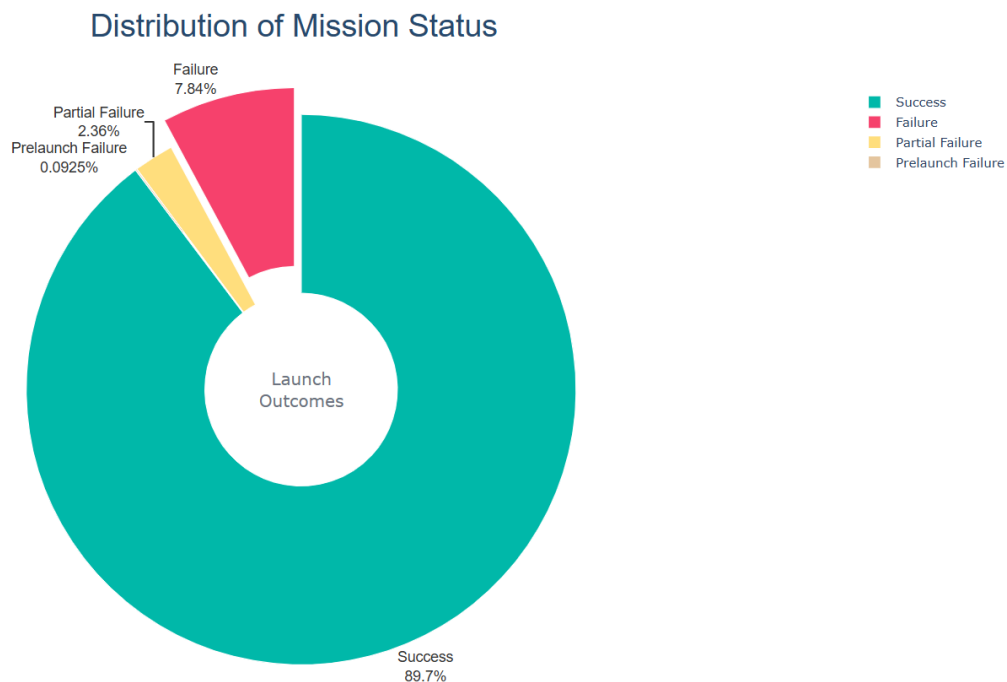
```
fig = go.Figure(data=[go.Pie(
    labels=df_by_mission_status['Mission_Status'],
    values=df_by_mission_status['Count'],
    hole=0.35,
    pull=pull_values,
    marker=dict(colors=colors, line=dict(color='FFFFFF', width=1)),
    textinfo='percent+label',
    textposition='outside',
    textfont=dict(size=14, family="Arial, sans-serif", color='#333333'),
    hoverinfo='label+value+percent'
)])
```

```
# 4. Update the layout for a clean, editorial look
```

```
fig.update_layout(
    title=dict(
        text='Distribution of Mission Status',
        font=dict(size=30, family="Arial, sans-serif", color='#27496D'),
        x=0.5, y=0.97
    ),
    annotations=[
        dict(
            text='Launch<br>Outcomes',
            x=0.5, y=0.5, font_size=16, font_color='#6B7280',
            showarrow=False
        )
    ],
    # showLegend=False, # Removing the Legend makes it cleaner
    margin=dict(t=90, b=40, l=40, r=40),
    paper_bgcolor='white',
```

```
plot_bgcolor='white'  
)  
  
fig.show()
```

Graph:



Interactive Graph Link: [Distribution of Mission Status](#)

The graph illustrates the remarkable reliability achieved in modern spaceflight: nearly 90% of all documented missions have been successfully completed. This is the result of decades of technical improvements, better testing procedures, reusable rocket technologies, and advances in navigation and safety systems that have been developed since the early days of the space race.

One particularly interesting detail is the extremely low number of pre-launch failures (only four missions), which are generally catastrophic incidents occurring before launch, such as fuel explosions or accidents on the launch pad—rare occurrences that highlight just how much modern space companies prioritize ground safety protocols and pre-launch verification systems long before the engines are ignited.

Analysis of Rocket Operational Status

In this section, we are going to examine the operational status of the rockets in the dataset to determine how many launch vehicles are still in service and how many have already been decommissioned.

This analysis illustrates the condition of rocket technology over time and provides insight into how the aerospace industry is continuously replacing older systems with newer, more efficient launch vehicles.

Calculating Rocket Status Distribution

The dataset is grouped by launch vehicle status using the Pandas `.groupby()` function, and the total number of entries in each category is counted to determine the distribution of active and retired launch vehicles.

```
df_by_rocket_status = (  
    df.groupby('Rocket_Status', as_index=False)  
    .size()  
    .rename(columns={'size': 'Count'})  
)
```

Creating Pie Chart

code:

```
color_map = {'StatusActive': '#08CB00', 'StatusRetired': '#D32626'}  
  
colors = [color_map.get(status, '#CCCCCC') for status in  
df_by_rocket_status['Rocket_Status']]  
  
fig = go.Figure(data=[go.Pie(  
    labels=df_by_rocket_status.Rocket_Status,  
    values=df_by_rocket_status.Count,  
    hole=0.4, # Creates a donut effect  
    marker=dict(colors=colors, line=dict(color='#FFFFFF', width=1)),  
    textinfo = 'percent+label',  
    textposition = 'outside',  
    textfont = dict(size=14, family="Arial, sans-serif", color='#333333'),  
    hoverinfo = 'label+value+percent'  
)])
```

```

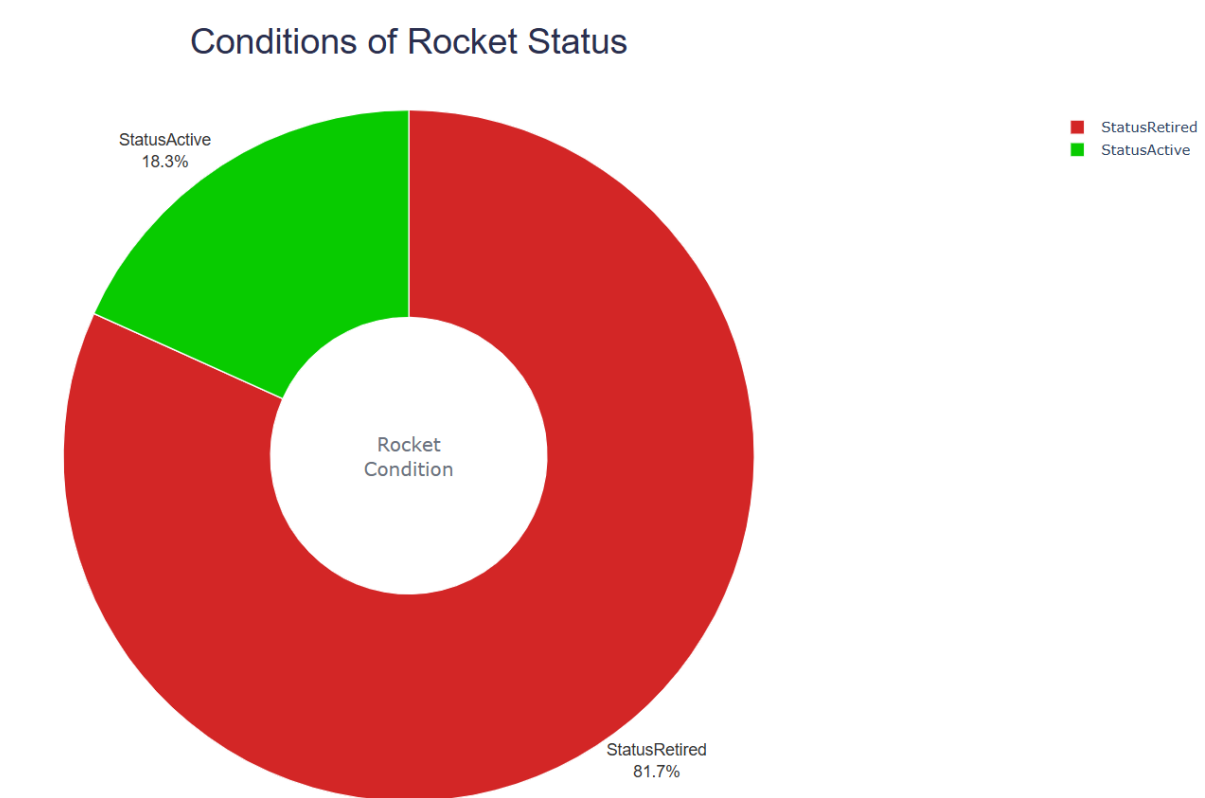
fig.update_traces(textinfo='percent+label')

fig.update_layout(
    title=dict(
        text='Conditions of Rocket Status',
        font=dict(size=30, family="Arial, sans-serif", color='#2A2F4F'),
        x=0.5,y=0.97
    ),
    annotations=[dict(text='Rocket<br>Condition',x=0.5, y=0.5, font_size=16,
font_color='#6B7280',showarrow=False )],
    margin=dict(t=90, b=40, l=40, r=40),
    paper_bgcolor='white',
    plot_bgcolor='white'
)

fig.show()

```

Graph:



Graph Link: [Conditions of Rocket Status](#)

The graph shows that the vast majority of rockets are decommissioned orbital launch systems: there are 3,534 retired rockets compared to just 789 that are still in service as 2020. This illustrates how rapidly aerospace technology is advancing and how frequently older rocket families are being replaced by more efficient designs.

This trend is closely linked to the long historical span of the dataset, as many legendary rockets from the Cold War era—such as the Saturn V and early Soviet launch vehicles—were developed for very specific political and technological objectives before eventually becoming obsolete or economically unviable.

Another key interpretation is that the relatively small proportion of active rockets could in fact point to increasing technological consolidation in the modern space industry, where reusable and versatile launch vehicles such as the Falcon 9 can carry out missions that previously required several different rocket families.

Global Distribution of Space Launches by Country

Space exploration has always been closely linked to geopolitical power, technological capabilities, and economic investment, which is why geographic analysis is essential to understanding the development of the global space industry. In this section, we are going to explore how launch activities are distributed across different countries by visualizing the total number of space missions on a world map.

Using a choropleth map, we can identify the leading spacefaring nations, observe regional patterns in launch activities.

Aggregating Launch Activity by Country

The dataset is grouped by both country and ISO code using the pandas `.groupby()` function to determine the total number of missions launched by each country. The aggregated values are then sorted in descending order.

```
# Aggregate mission counts by location
df_country = (df.groupby(['Country', 'ISO'], as_index=False)
               .agg(Total_Missions=('Mission_Status', 'count')))

# Sort the dataframe for better visualization
df_country.sort_values('Total_Missions', ascending=False, inplace=True)
```

Creating the Choropleth Visualization

Using Plotly Express, a choropleth map is created to visually represent launch activity by region based on color intensity, with darker areas indicating a higher number of missions.

By combining ISO country codes, interactive tooltips, and a continuous color scale, the visualization offers an intuitive and engaging way to explore the global distribution of space launches.

code:

```
# Plotting the Choropleth
world_map = px.choropleth(
    df_country,
    locations='ISO',
    color='Total_Missions',
    hover_name='Country',
    custom_data=['ISO', 'Total_Missions'],
    color_continuous_scale='matter',
    title='<b style="color:#152346;">Global Rocket Launches by Country</b>'
)

# Layout adjustments
world_map.update_layout(
    font_family='Arial, Helvetica, sans-serif',
    font_color='#2c3e50',
    title_font_size=24,
    title_font_color='#922062',
    title_x=0.03,
    margin=dict(t=80, l=0, r=0, b=0),
    geo=dict(
        showframe=False, # remove the border around the world
        showcoastlines=False, # remove the thick lines
        projection_type='natural earth', # an earth projection
        bgcolor='rgba(0,0,0,0)', # transparent background
        lakecolor='ffffff', # fill lakes with white to blend in cleanly
    ),

    coloraxis_colorbar=dict(
        title=dict(text="Total Launches", font=dict(size=14, color='#2c3e50')),
        thicknessmode="pixels", thickness=12,
        lenmode="pixels", len=250,
        yanchor="middle", y=0.5,
        xanchor="left", x=0.95,
        tickvals=[0, 1, 2],
        ticktext=['1', '10', '100+'],
        tickfont=dict(size=12, color='#2c3e50')
    )
)
```

```

)

world_map.update_traces(
    marker_line_color='#ffffff', # a subtle white borders between countries
    marker_line_width=0.4, # but they must be thin
    # Hover arrangements:
    hovertemplate=(
        '<b style="font-size: 14px; color: #152346;">{%{hovertext}</b><br>' +
        '<span style="font-size: 13px; color: #555;">ISO: </span>' +
        '<b style="font-size: 14px; color: #2c3e50;">{%{customdata[0]}</b><br>' +
        '<span style="font-size: 13px; color: #555;">Total Launches: </span>' +
        '<b style="font-size: 14px; color: #152346;">{%{customdata[1]}</b><br>' +
        '<extra></extra>'
    ),
    hoverlabel=dict(bgcolor="white", font_size=14, font_family="Arial",
bordercolor="#e2e8f0")
)

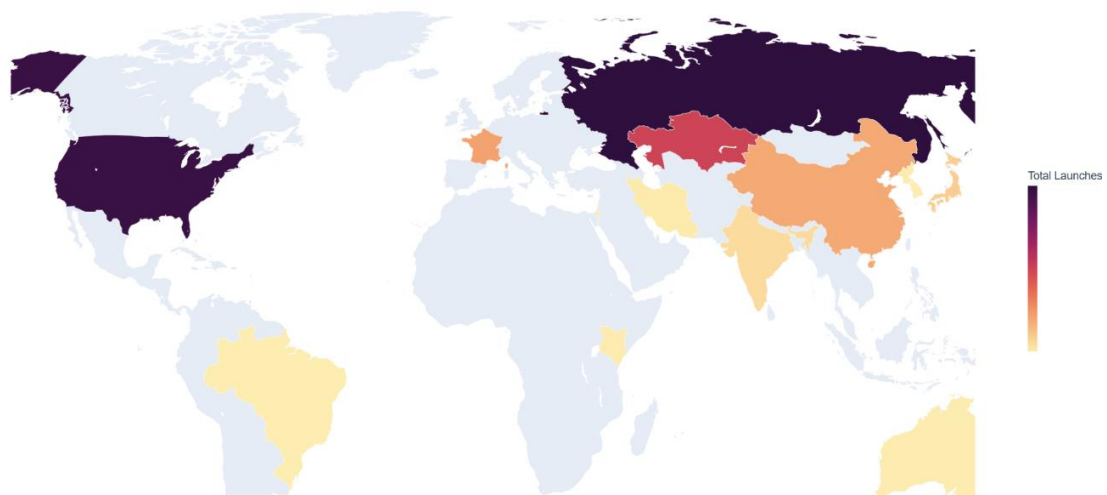
# world_map.write_html("interactive_graphs/space_race/world_map_by_launches.html")

world_map.show()

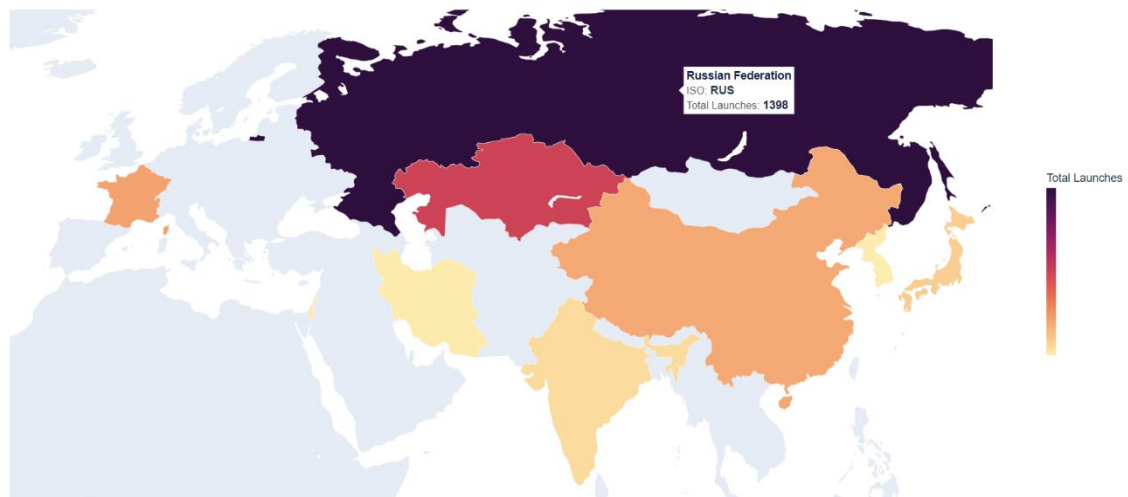
```

Graph:

Global Rocket Launches by Country



Global Rocket Launches by Country



Interactive Graph Link: [Global Rocket Launches by Country](#)

The choropleth map illustrates the historical dominance of the Russian Federation and the United States in global space exploration—a pattern largely shaped by decades of technological competition during the Space Race, during which both nations invested enormous financial and scientific resources in launch infrastructure and rocket development.

Another notable observation is the exceptionally high number of launches from Kazakhstan, which is primarily attributable to the historic Baikonur Cosmodrome—the world’s first and largest operational space launch facility, which was used extensively for Soviet and later Russian missions.

The chart also reflects the ongoing globalization of space exploration, with countries such as China, India, and New Zealand playing an increasingly significant role. This suggests that the future of the space industry may be more broadly distributed internationally, rather than being dominated by just a few superpowers.

Country-Level Launch Success Rate Analysis

While the total number of launches indicates which countries have been most active in space exploration, this does not necessarily say anything about operational reliability or technological capabilities.

In this section, we are going to analyze launch success rates by country to determine which nations have achieved the highest mission reliability relative to their total number of launches.

This approach provides a more balanced picture of performance in the aerospace sector, as it takes into account not only how often countries have launched missions, but also how consistently they have been able to complete them successfully.

Filtering and Aggregating Failure Data

The dataset is first filtered using the `query()` function in pandas to identify all failed missions, taking into account both complete and partial failures.

These filtered records are then grouped by country and aggregated using the `count()` function to calculate the total number of failed missions for each country.

```
# Here we filter all the entries where mission status is not success
df_failure = df.query("Mission_Status != 'Success'")

# We group this entries by country to get desirable entries
df_failure = df_failure.groupby(["Country"],
as_index=False).agg(Total_Failures=("Mission_Status", "count"))
```

Combining Mission and Failure Statistics

The aggregated dataset on failed launches is merged with the mission-specific dataset at the country level we formed at the previous step using the `merge()` function in pandas, so that the total number of launches and the total number of failed launches can be analyzed together in a single DataFrame.

Missing values are replaced with zeros using `fillna()` to ensure that countries with no recorded failed launches are still correctly included in the analysis.

```
# Adding an additional column for failures to df_country database.
df_success_vs_failure = df_country.merge(df_failure, on="Country",
how="left").sort_values('Total_Failures', ascending=False)

# In cast there were no failure
df_success_vs_failure.fillna(0, inplace=True)
```

Combining the total number of missions with the total number of failures yields a new metric called *Success_Rate*, which allows for a percentage-based comparison between countries rather than relying solely on launch figures.

```
# # Creating a new column for failure percentages:
df_success_vs_failure["Success_Rate"] = (df_success_vs_failure.Total_Missions -
df_success_vs_failure.Total_Failures) / df_success_vs_failure.Total_Missions * 100

# print("\nMost Successful Countries by Launch Success Rate:")
df_success_vs_failure.sort_values('Success_Rate', ascending=False, inplace=True)
```

Visualizing Success Rates with a Bar Chart

code:

```
bar = px.bar(
    df_success_vs_failure[:10],
    x='Country',
    y='Success_Rate',
    color='Country',
    custom_data=['Total_Missions', 'Total_Failures', 'Success_Rate'],
    color_discrete_sequence=px.colors.qualitative.Bold,
    text='Success_Rate', # Display the success rates
)

# Options for styling, text formatting, and hover template
bar.update_traces(
    texttemplate='<b>{text:,.2f}%</b>', # Formatting rates to percentages.
    textposition='outside', textfont_size=12, marker_line_width=0, # bars.
    # These are the arrangements for hover display
    hovertemplate=(
        "<b>{x}</b><br>" +
        "Total Missions: %{customdata[0]}<br>" +
        "Total Failures: %{customdata[1]:,.0f}<br>" +
        "Success Rate: %{customdata[2]:,.0f}" +
        "<extra></extra>"
    )
)

bar.update_layout(
    template='plotly_white',
    title=dict(text='<b>Most Successful Countries by Space Launch Success
Rate</b>'),
```

```

        font=dict(size=26, family='Arial, Helvetica, sans-serif',
color='#008695'), x=0.025),
    xaxis_title='',
    yaxis_title='Percentages',
    yaxis_title_font_size=16,
    yaxis_title_font_color="#008695",

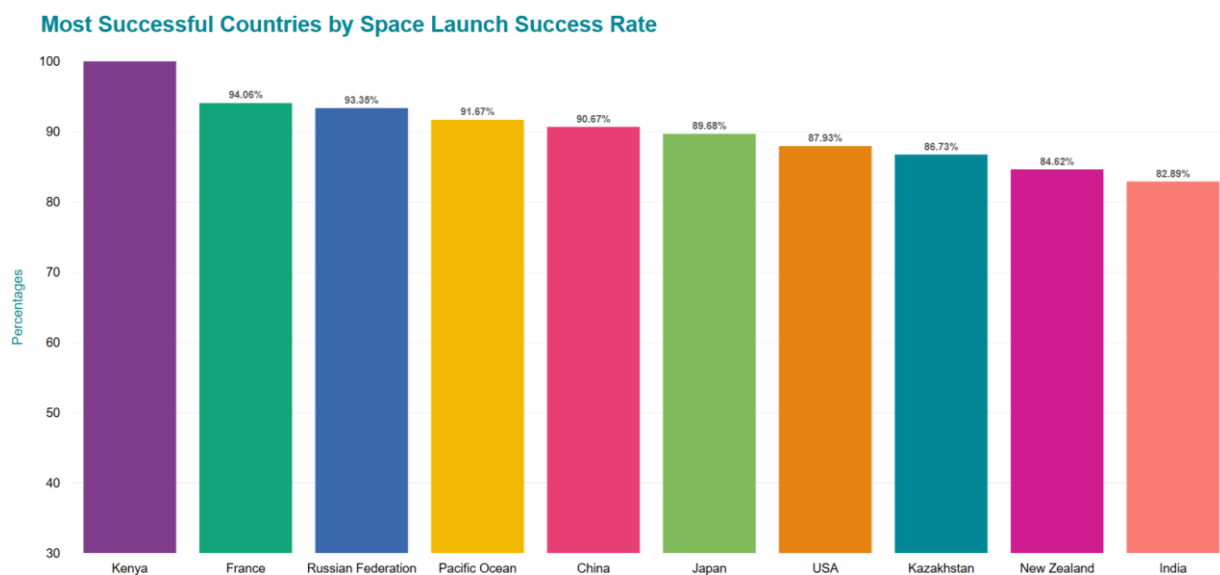
    yaxis_range=[30, 100],
    font=dict(family='Arial, sans-serif', size=13, color='#555555'),

    xaxis=dict(showgrid=False, ticklabelposition="outside", ticklabelstandoff=7,
    tickfont=dict(size=15, family="Arial", color='black')),

    yaxis=dict(showgrid=True, showticklabels=True, zeroline=False,
    ticklabelposition="outside", ticklabelstandoff=13,
    tickfont=dict(size=15, family="Arial", color='black')),
    showlegend=False, # No legend
    coloraxis_showscale=False, # No coloraxis
    margin=dict(l=10, r=70, t=70, b=20),
)
bar.show()

```

Graph:



Interactive Graph Link: [Most Successful Countries by Space Launch Success Rate](#)

One of the most striking findings in the graph is Kenya's 100% success rate; however, this result should be interpreted with caution, as it is based on only 9 launches, most of which were associated with the historic San Marco launch Platform, which was operated primarily by Italian and European space programs.

France also boasts an exceptionally high success rate, which is primarily due to the activities of Arianespace and the famous space center in French Guiana, which—thanks to its favorable location on the equator, which reduces fuel requirements for orbital missions—has become one of the most reliable launch sites in the world.

Another key finding is that countries with extremely high launch rates, such as the United States and the Russian Federation, have naturally experienced more failures, as they have conducted large-scale test programs over many decades. This suggests that a high volume of launches is often accompanied by a greater willingness to take technological risks and a higher degree of innovation.

Countries with the Highest Failure Rates

In this section, we are going to inspect the countries with the lowest launch rates to better understand the technological, economic, and historical challenges faced by emerging or less experienced space programs.

Again, this analysis focuses on reliability ratios, thereby providing insight into how challenging and risky early-stage rocket development can be for countries with limited aerospace infrastructure or smaller test programs.

Most Unsuccessful Countries by Launch Success Rate:

```
# Most Unsuccessful Countries
print("\nMost Unsuccessful Countries by Launch Success Rate:")
df_success_vs_failure.sort_values('Success_Rate', ascending=True, inplace=True)
print(df_success_vs_failure.head())
```

	Country	ISO	Total_Missions	Total_Failures	Success_Rate
14	Brazil	BRA	3	3	0.00
15	South Korea	KOR	3	2	33.33
8	Iran	IRN	14	9	35.71
13	North Korea	PRK	5	3	40.00
12	Australia	AUS	6	3	50.00

The results show that some countries with the lowest success rates also have an extremely small total number of launches, which means that even a few failed missions have a dramatic impact on their success rates.

Brazil stands out in particular because of the tragic explosion at the Alcântara Space Center in 2003, in which aerospace personnel lost their lives before the launch, significantly hampering the country's space ambitions for years to come.

Countries such as Iran and North Korea also have lower success rates, likely due to the technical challenges involved in developing launch vehicles—challenges that are worsened by economic constraints, limited testing capabilities, and international sanctions—factors that can significantly slow progress in the aerospace sector.

Hierarchical Analysis of Countries, Organizations, and Mission Success

In this section, we are going to conduct a multi-level analysis of global space activities by simultaneously examining the relationships between countries, organizations, and mission outcomes.

Using a sunburst chart, we can examine the hierarchical distribution of launch activities and thus determine which organizations in each country play a leading role and how successful their missions have been over time. This visualization provides a deeper understanding of the structure of the global space industry by combining geographic factors, organizational activities, and mission reliability into a single interactive analysis.

Aggregating Launch Activity Across Multiple Dimensions

The dataset is grouped by country, organization, and mission status using the Pandas `groupby()` function to create a hierarchical structure suitable for the Sunburst visualization. The grouped datasets are then aggregated by counting the number of launches, allowing the analysis to determine how mission outcomes differ across different organizations and countries.

```
# Grouping and Sorting
country_organisation = (df.groupby(['Country', 'Organisation', 'Mission_Status'],
as_index=False)
                        .agg(Launch_Number=('Rocket_Status', 'count')))

country_organisation.sort_values('Launch_Number', ascending=False)
```

Creating the Sunburst Visualization

Using Plotly Express, a sunburst chart is created to illustrate the hierarchical relationships between countries, organizations, and mission outcomes through nested concentric circles.

Mission statuses are color-coded using a custom color scheme to visually distinguish between successful, failed, and partially failed launches.

code:

```
burst = px.sunburst(
    country_organisation,
    path=['Country', 'Organisation', 'Mission_Status'],
    values='Launch_Number',
    template="plotly_white"
)

status_colors = {
    "Success": "#4CAF50",
    "Failure": "#E53935",
    "Partial Failure": "#FB8C00",
    "Prelaunch Failure": "#1E88E5",
    "Unknown": "#9E9E9E",
}

sequential_palette = px.colors.qualitative.Pastel

# We are going to customize the coloring of the labels here
# Extract the labels generated by Plotly and assign our custom colors to them one
# by one
labels = burst.data[0].labels
custom_colors = []
seq_index = 0

for label in labels:
    if label in status_colors:
        custom_colors.append(status_colors[label])
    else:
        custom_colors.append(sequential_palette[seq_index %
len(sequential_palette)])
        seq_index += 1

burst.update_traces(
    marker=dict(colors=custom_colors, line=dict(color='FFFFFF', width=1.5)),
    textfont=dict(family="Arial, sans-serif", color="#2C3E50"),
    hovertemplate="<b>{%label}</b><br>Launches: {%value}<extra></extra>"
)
```

```

burst.update_layout(
    margin=dict(t=70, l=10, r=10, b=10),
    title=dict(text='<b>Rocket Launches by Country, Organisation, and Mission Status</b>',
               font=dict(size=22, family='Arial, Helvetica, sans-serif',
                           color='#2C3E50'), x=0.5),
    uniformtext=dict(minsize=11, mode="hide"),
)

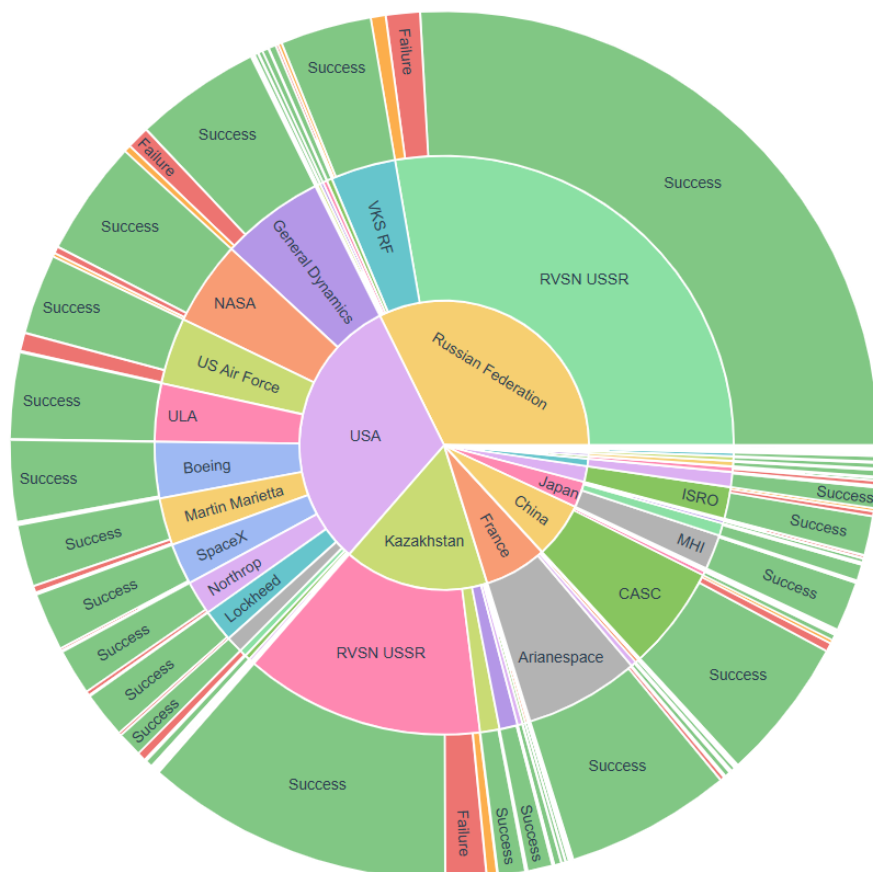
#
burst.write_html("interactive_graphs/space_race/sunburst_all_organizations.html")

burst.show()

```

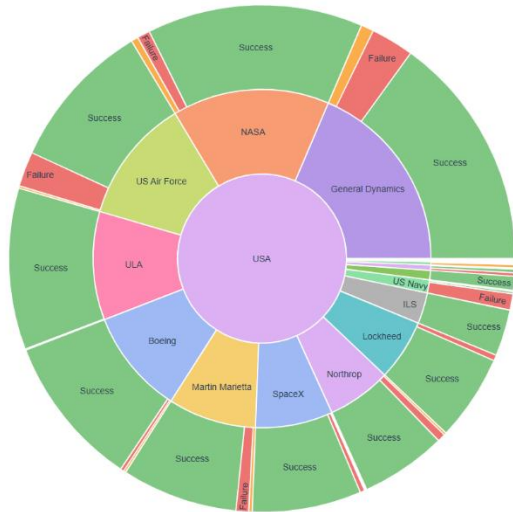
Graph:

Rocket Launches by Country, Organisation, and Mission Status

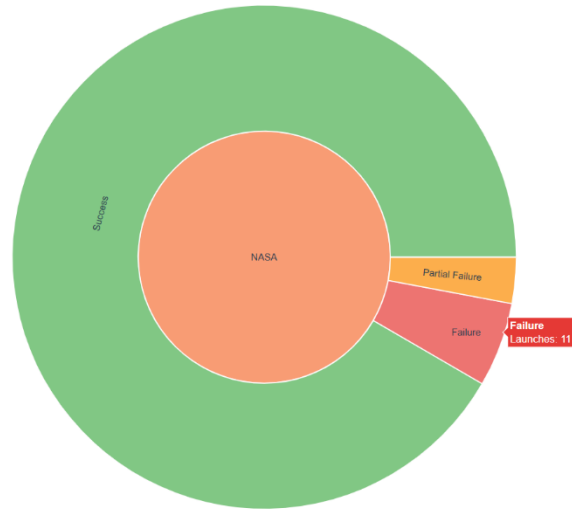


Selecting USA:

Rocket Launches by Country, Organisation, and Mission Status



Rocket Launches by Country, Organisation, and Mission Status



Interactive Graph link: [Rocket Launches by Country, Organization, and Mission Status](#)

The Sunburst chart shows that the global space industry is heavily concentrated among a small number of dominant organizations, with the Soviet-era RVSN of the USSR dominating launch activities not only in the Russian Federation but also in Kazakhstan —this is due to the historic Baikonur Cosmodrome, which served for decades as the operational hub of Soviet space missions.

In contrast, the United States has a far more decentralized structure, in which organizations such as NASA, Boeing, SpaceX, and Lockheed share the workload for launch operations. This reflects a competitive ecosystem in which both government agencies and private companies coexist while driving innovation.

Another key finding is that even countries such as France, China, and Japan are heavily dominated by one or two major organizations, suggesting that it is extremely difficult to maintain multiple independent aerospace organizations, as the space industry involves enormous financial costs, technological hurdles, high infrastructure requirements, and long-term research investments.

Analysis of Organizational Spending on Space Missions

In the previous sections, we analyzed organizations based on the number of their launches; in this section, we are going to shift our focus to total spending on space missions.

Examining financial expenditures offers a different perspective on the space industry and helps us understand which organizations have invested the most resources in launch operations, technological development, and large-scale aerospace programs.

Aggregating Launch Counts and Total Mission Spending

Using the pandas functions `.groupby()` and `.agg()`, the dataset is aggregated by organization to calculate both the total recorded expenditures and the total number of launches conducted by each organization.

This approach allows us to compare the scale of activities and financial investments simultaneously.

```
# Getting the total number of launches and total spending
org_price_with_na = df.groupby(['Organisation'], as_index=False).agg(
    Total_Spending=("Price", "sum"),
    Launch_Count_Total=("Mission_Status", "count")
)
```

When calculating total expenditures, rows with missing price data are included because pandas automatically ignores NaN values when summing, ensuring that the missing entries do not skew the total expenditures.

However, the total number of launches still includes all missions, regardless of whether cost data is available. Therefore, it is important to determine separately how many launches actually contribute to the recorded expenditure data.

To improve the reliability of the financial analysis, we also determine the number of launches for which price information is available.

Since only these missions (the ones that have cost values) contribute directly to total expenditures, this additional metric allows for a clearer interpretation of the organization's spending habits and prevents misleading comparisons.

We are going to also plot the total number of Launches which don't have na values in price and makes up this general total cost.

Filtering Missing Price Data and Recalculating Launch Counts

The dataset is filtered using the `dropna()` function to remove entries with missing price data, resulting in a separate DataFrame that contains only missions with available expenditure data.

The filtered data is then grouped and recounted to determine how many launches per organization are actually included in the expenditure analysis.

```
# for no price data, we have to create another column with dropping the na values
prices_without_na = df.dropna(subset=['Price']).copy()

org_price_without_na = prices_without_na.groupby(['Organisation'], as_index=False)
                                .agg(Launch_Count_Spending_Accessible=(
                                    "Mission_Status", "count"))
```

Combining Financial and Launch Data

The two aggregated DataFrames are merged using the pandas `.merge()` function, allowing the number of runs, the available output data, and the total output to be analyzed together in a single structured dataset.

Missing values resulting from the merge are replaced and converted to integer format to ensure numerical consistency and improve the readability of the data.

```
# Merging two dataframe we have:
org_price_with_na = org_price_with_na.merge(org_price_without_na,
on="Organisation", how="left")

# Converting second values to integers
# Since it can include na values they are held on float numbers
org_price_with_na["Launch_Count_Spending_Accessible"] = (
    org_price_with_na["Launch_Count_Spending_Accessible"]
    .fillna(0)
    .astype(int)
)
```

Calculating Average Spending per Launch and Ranking Organizations

To enable a comparison of average spending per mission across organizations, a new metric called *Avr_Spending_by_Launch* is calculated by dividing the organization's total spending by the number of launches for which price data is available.

Finally, the dataset is sorted in descending order by total expenditure per mission, and the 20 organizations with the highest expenditure are selected for further visualization and analysis.

```
# Adding a ew column to display the average spending by Launch:
org_price_with_na["Avr_Spending_by_Launch"] =
    Total_Spending / Launch_Count_Spending_Accessible

# Sorting by Total Spending:
org_price_with_na = org_price_with_na.sort_values('Total_Spending',
ascending=False)

# Take the most 20 Organizations
org_price_with_na_top_20 = org_price_with_na[:20]
```

Formatting Financial Metrics for Visualization

To improve readability and presentation quality, a custom formatting function is created that converts large numerical values into more intuitive financial terms such as millions (M) and billions (B).

This type of preprocessing is frequently used in data visualization to make large volumes of financial data easier to interpret and visually compare, without overwhelming the reader with long strings of numbers.

```
# We define a function to display the money values nicely
def format_spending_label(val_in_millions):
    if val_in_millions < 1000:
        return f"${val_in_millions:,.0f}M" # 500.00 -> $500M
    else:
        return f"${val_in_millions / 1000:,.1f}B" # 2500.00 -> $2.5B
```

Applying Financial Label Formatting

Custom formatting is applied to the Total Expenditures column to make the labels clearer, while the original expenditure values are also converted from millions to billions to ensure a more intuitive scale for the display.

```
org_price_with_na_top_20['Display_Label'] =  
org_price_with_na_top_20['Total_Spending'].apply(format_spending_label)  
  
# Since values are in millions, we are going to need formatting:  
org_price_with_na_top_20['Total_Spending_B'] =  
org_price_with_na_top_20['Total_Spending'] / 1000
```

Creating the Organizational Spending Bar Chart

Using Plotly Express, a horizontal bar chart is created to compare the total expenditures of the top 20 organizations in the dataset.

Additional interactive metadata, such as the number of launches, average spending per mission, and the total number of available spending records, is incorporated via custom hover data, allowing the visualization to provide both financial and operational insights simultaneously.

code:

```
c_bar = px.bar(  
    org_price_with_na_top_20,  
    x='Total_Spending_B',  
    y='Organisation',  
    orientation='h',  
    custom_data=[  
        'Total_Spending_B',  
        'Launch_Count_Spending_Accessible',  
        'Avr_Spending_by_Launch',  
        'Launch_Count_Total'  
    ],  
    text='Display_Label',  
    color=np.log10(org_price_with_na_top_20['Total_Spending_B']),  
    color_continuous_scale='Deep',  
)  
  
c_bar.update_traces(  

```

```

    texttemplate='<b>{%text}</b>',
    textposition='outside', textfont=dict(size=12, color='#2C3E50'),
    marker_line_width=0,
    hovertemplate=(
        "<b style='font-size:15px;'>{%y}</b><br><br>" +
        "<span>Total Spending:</span> <b>${customdata[0]:,.2f} Billion</b><br>" +
        "<span>Number of Launches accounts for this Amount:</span>"
    )
    <b>{%customdata[1]:,.0f}</b><br>" +
        "<span>Average Cost per Launch:</span> <b>${customdata[2]:,.0f}M</b><br>" +
        "<span>Total Number of Launches (General):</span>"
    <b>{%customdata[3]:,.0f}</b>" +
        "<extra></extra>" # Removes the messy secondary trace box
    )
)

c_bar.update_layout(
    template='plotly_white',
    title=dict(
        text='<b>Top 20 Organizations by Total Launch Expenditure</b>',
        font=dict(size=25, family='Arial, sans-serif', color='#3f4b86'),
        x=0.03, xanchor='left'
    ),
    xaxis_title=dict(
        text='<b>Total Expenditure on Launches</b>',
        font=dict(size=14, family='Arial, sans-serif', color='#3f4b86')),
    yaxis_title='',
    font=dict(family='Helvetica Neue, sans-serif', size=13, color='#5F6B78'),
    showlegend=False,
    coloraxis_showscale=False,
    margin=dict(l=10, r=70, t=80, b=30),
    plot_bgcolor='rgba(0,0,0,0)',
    paper_bgcolor='rgba(0,0,0,0)'
)

c_bar.update_xaxes(
    showgrid=True, gridcolor='#F0F2F6', zeroline=False,
    ticklabelposition="outside", ticklabelstandoff=7,
    tickfont=dict(size=13, color='#7f808f'), type='log',
    tickvals=[0.01, 0.1, 1, 10, 100], # specific tick locations for labels
    ticktext=['$10M', '$100M', '$1B', '$10B', '$100B']
)

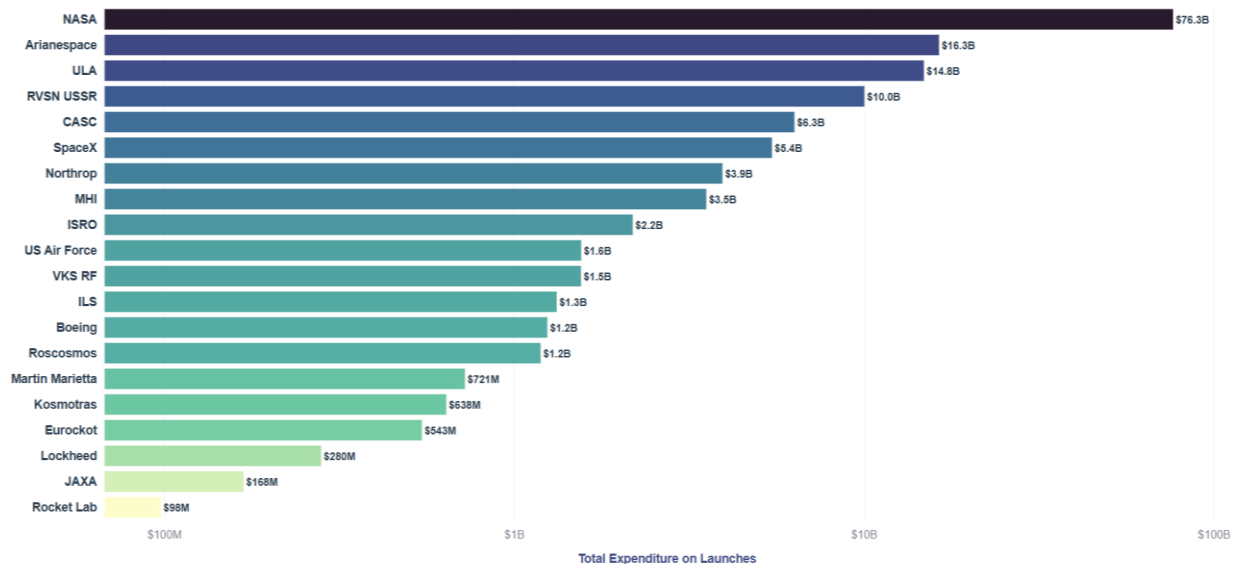
c_bar.update_yaxes(
    showgrid=False, showticklabels=True, zeroline=False,
    ticklabelstandoff=8, tickfont=dict(size=14, color='#2C3E50', weight='bold'),
    categoryorder='total ascending'
)

c_bar.show()

```

Graph:

Top 20 Organizations by Total Launch Expenditure



Interactive Graph Link: [Top 20 Organizations by Total Launch Expenditure](#)

The chart illustrates NASA's enormous financial dominance, with total recorded expenditures exceeding \$76 billion—a reflection of decades of investment in historically ambitious programs, space exploration, and cutting-edge research.

An important detail evident from the graph is that for some organizations, particularly the RVSN of the USSR, very few publicly available cost records exist, even though they recorded an extremely high number of launches. This reflects the secrecy that surrounded many Soviet-era aerospace and military programs during the Cold War.

Furthermore, when we hover over the bars it becomes clear that some organizations have unusually high average expenditure per launch, as publicly available cost data is only available for a small fraction of their missions.

This means that the expenditure analysis reflects both actual investment patterns and differences in financial transparency among governments, military agencies, and private aerospace companies.

Identifying the Most Expensive Space Launches

In this section, we are going to analyze the launches with the highest documented mission costs to better understand which space programs required the largest financial investments.

Examining these outliers highlights how ambitious technological goals, political competition, and long-term research programs have significantly influenced the cost-effectiveness of space exploration.

```
df_top10_launches = df.sort_values("Price", ascending=False)
print("\n Most Expensive Launches")
print(df_top10_launches.head(6))
```

Most Expensive Launches:

Organisation	Location	Date	Detail	Rocket_Status	Price	Mission_Status	Country
RVSN USSR	Site 110/37, Baikonur Cosmodrome...	1988-11-15	Energiya Buran	StatusRetired	5,000.00	Success	Kazakhstan
RVSN USSR	Site 250, Baikonur Cosmodrome...	1987-05-15	Energiya Polyus Space Station	StatusRetired	5,000.00	Success	Kazakhstan
NASA	LC-39A, Kennedy Space Center...	1968-04-04	Saturn V Apollo 6	StatusRetired	1,160.00	Partial Failure	USA
NASA	LC-39A, Kennedy Space Center...	1968-12-21	Saturn V Apollo 8	StatusRetired	1,160.00	Success	USA
NASA	LC-39A, Kennedy Space Center...	1967-11-09	Saturn V Apollo 4	StatusRetired	1,160.00	Success	USA
NASA	LC-39A, Kennedy Space Center...	1973-05-14	Saturn V Skylab 1	StatusRetired	1,160.00	Success	USA

The results show that the Soviet “Energiya” program tops the list with exceptionally high costs per launch(\$5 Billion each), primarily because the program’s total development costs were spread across just two launches before the collapse of the Soviet Union forced the project to be shut down.

Another notable observation is the strong presence of NASA missions using the Saturn V rocket, reflecting the enormous technical complexity and political significance of the Apollo program, in which financial efficiency often had to take a back seat to technological achievements and national prestige during the space race.

It is also important to note that the dataset calculates launch costs by allocating the estimated total development costs of a rocket program across the number of launches conducted under that program.

This explains why launch vehicles with only a few missions, such as “Energiya,” appear to be significantly more expensive per launch.

Analysis of Space Launch Programs and Rocket Families

In this section, we are going to analyze the most important space launch programs, represented by the *Rocket_Type* attribute, which describes the families of launch vehicles that have been used throughout the history of space exploration.

These rocket programs can be considered the “launch systems” of the space industry, as they are responsible for transporting satellites, spacecraft, space stations, and scientific missions into orbit or into space.

Since the financial data in the dataset is primarily linked to these launch vehicle programs, analyzing it provides deeper insights into how different countries and organizations have allocated their resources across long-term space projects and technological generations

Aggregating Program Expenditure and Launch Activity

The dataset is first filtered using `.dropna()` function to remove entries with missing price data, to include only the launches with available cost in the financial analysis.

Using the `.groupby()` and aggregation functions, the data is then grouped by rocket type, country, and organization to calculate the total expenditure, the total number of launches, and the average expenditure per launch for each space program.

```
prices_without_na = df.dropna(subset=['Price']).copy()

df_program = prices_without_na
    .groupby(['Rocket_Type', 'Country', 'Organisation'], as_index=False)
    .agg(
        Total_Spending=("Price", "sum"),
        Launch_Count_Total=("Mission_Status", "count"),
        Avr_Spending_by_Launch=("Price", "mean"),
    )
```

Ranking and Filtering the Largest Space Programs

The aggregated dataset is sorted in descending order by total expenditure to identify the most financially significant rocket programs in the history of spaceflight. The 20 most significant launch vehicle programs are then selected for visualization and further comparative analysis.


```
# sorting the Programs:
df_program = df_program.sort_values("Total_Spending", ascending=False)

# We are going to plot most 20 Space Launch Program by Expenditure
df_program_top20 = df_program[:20]
```

Applying Financial Display Formatting

Previously defined formatting functions are reused to convert large numerical output values into more readable financial figures expressed in millions and billions of dollars.

```
# We are going to use same value formatting as before:
df_program_top20['Display_Label'] = df_program_top20['Total_Spending']
                                   .apply(format_spending_label)

# Since values are in millions, we are going to need formatting:
df_program_top20['Total_Spending_B'] = df_program_top20['Total_Spending'] / 1000
```

Creating the Bar Graph for Program Expenditures

Plotly Express is used to create a horizontal bar chart that compares the total expenditures of the largest rocket programs while displaying the number of launches and average mission costs via interactive hover data.

code:

```
c_bar = px.bar(
    df_program_top20,
    x='Total_Spending_B',
    y='Rocket_Type',
    orientation='h',
    custom_data=[
        'Total_Spending_B',
        'Launch_Count_Total',
        'Avr_Spending_by_Launch',
    ],
    text='Display_Label',
    color=np.log10(df_program_top20['Total_Spending_B']),
    color_continuous_scale='Emrld',
)
```

```

c_bar.update_traces(
    texttemplate='<b>{%text}</b>', textposition='outside',
    textfont=dict(size=12, color='#2C3E50'),
    marker_line_width=0,
    hovertemplate=(
        "<b style='font-size:15px;'>{%y}</b><br><br>" +
        "<span>Total Spending:</span> <b>${customdata[0]:,.2f} Billion</b><br>" +
        "<span>Number of Launches</span> <b>${customdata[1]:,.0f}</b><br>" +
        "<span>Average Cost per Launch:</span> <b>${customdata[2]:,.0f}M</b><br>" +
        "<extra></extra>"
    )
)

c_bar.update_layout(
    template='plotly_white',
    title=dict(
        text='<b>Top 20 Missile Program by Total Launch Expenditure</b>',
        font=dict(size=25, family='Arial, sans-serif', color='#54a484'),
        x=0.03, xanchor='left'
    ),
    xaxis_title=dict(
        text='<b>Total Expenditure on Launches</b>',
        font=dict(size=14, family='Arial, sans-serif', color='#54a484')),
    yaxis_title='',
    font=dict(family='Arial, sans-serif', size=13, color='#5F6B78'),
    showlegend=False,
    coloraxis_showscale=False,
    margin=dict(l=10, r=70, t=80, b=30),
    plot_bgcolor='rgba(0,0,0,0)',
    paper_bgcolor='rgba(0,0,0,0)'
)

c_bar.update_xaxes(
    showgrid=True, gridcolor='#F0F2F6', zeroline=False,
    ticklabelposition="outside", ticklabelstandoff=7,
    tickfont=dict(size=13, color='#0d505d'), type='log',
    tickvals=[2, 4, 10, 16], # specific tick locations for exact labels
    ticktext=['$2B', '$4B', '$10B', '$16B']
)

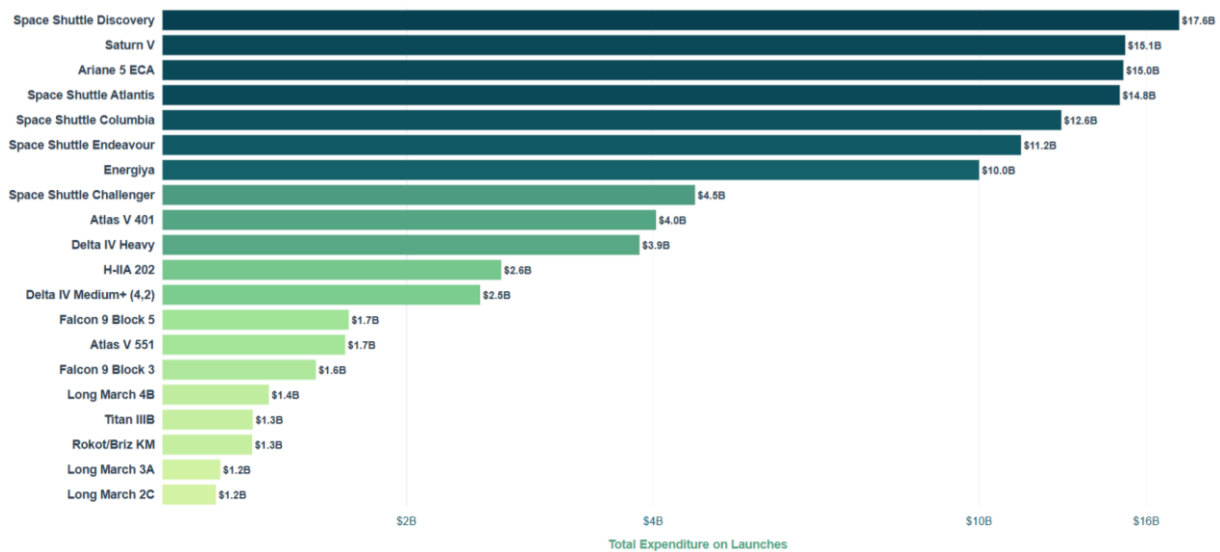
c_bar.update_yaxes(
    showgrid=False, showticklabels=True, zeroline=False,
    ticklabelstandoff=8, tickfont=dict(size=14, color='#2C3E50', weight='bold'),
    categoryorder='total ascending'
)

c_bar.show()

```

Graph:

Top 20 Missile Program by Total Launch Expenditure



Interactive Graph Link: [Top 20 Missile Program by Total Launch Expenditure](#)

The chart illustrates the enormous financial costs associated with reusable heavy-lift launch systems, with Space Shuttle orbiters such as Discovery, Atlantis, and Columbia accounting for the bulk of total expenditures due to their decades-long service life, costly maintenance requirements, and the technological complexity of reusable manned spaceflight systems.

Another notable observation is the continued presence of Cold War-era rockets such as Saturn V and Energiya at the top of the rankings, illustrating how lunar exploration and ambitions in the realm of extremely heavy-lift launch vehicles required unprecedented investments in technology that modern programs still rarely match.

At the same time, newer systems such as Falcon 9 and the Long March series appear significantly more cost-effective, suggesting that advances in reusable rocket technology, standardized manufacturing, and corporate competition are gradually breaking down the cost barriers to access to space.

Evolution of Average Launch Costs Over Time

In this section, we are going to investigate how the average cost per launch for major rocket programs has changed throughout the history of space exploration.

Rather than focusing solely on total expenditures, this analysis offers a perspective on cost-efficiency by examining how efficiently various launch systems have carried out missions over time and how technological advances have influenced the cost-efficiency of launches.

By presenting these trends chronologically, we can better understand the transition from the extremely expensive heavy-lift systems of the Cold War era to more modern launch vehicles.

Expanding the Dataset with Program Timeline Information

Since each rocket program consists of multiple launches spanning several years, a representative time value is needed to represent each program as a single point on the graph.

To this end, the dataset is grouped by rocket type and country, and the average launch date is calculated using Pandas aggregation functions, thereby determining an approximate “average operational duration” for each launch program.

This additional feature allows the financial dataset to be linked with program-level temporal information, enabling a chronological analysis of launch costs across decades of spaceflight history.

```
date_program = df.groupby(["Rocket_Type", "Country"])
                .agg(Avr_Date=("Date", "mean"))

df_program = df_program.merge(date_program, on="Rocket_Type")
```

Applying Financial Display Formatting

Previously defined formatting functions are reused to convert large numerical output values into more readable financial figures expressed in millions and billions of dollars.

```
df_program['Avr_Spending_by_Launch'].apply(format_spending_label)
df_program['Total_Spending'].apply(format_spending_label)

df_program['Avr_Spending_by_Launch_B'] = df_program['Avr_Spending_by_Launch'] / 1000
```

Creating the Scatter Plot with Trendline

Using Plotly Express, a scatter plot is created to compare the average costs of rocket programs with their average operational lifespans. A lowess-trendline is added to smooth out the fluctuations and reveal broader patterns in launch economics.

This illustrates how advances in engineering, manufacturing, and reusable rocket technology have influenced the cost of access to space over time.

code:

```
fig_px = px.scatter(
    df_program,
    x='Avr_Date',
    y='Avr_Spending_by_Launch_B',
    trendline='lowess',
    opacity=0.6,
    title='<b>Average Cost per Launch by Program Over Time</b>',
    custom_data=[
        'Rocket_Type',
        'Avr_Date',
        'Organisation',
        'Country',
        'Total_Spending_Formatted',
        'Launch_Count_Total',
        'Avr_Spending_Formatted'
    ],
    template='plotly_white',
)

hover_html = (
    "<b style='font-size:16px; color:#1A252F'>{%customdata[0]}</b><br>" +
    "<span style='font-size:13px; color:#5F6B78'>{%x|B %Y}</span><br>" +
    "<span style='font-size:13px; color:#5F6B78'>{%customdata[2]}</span><br>" +
    "<span style='font-size:13px; color:#5F6B78'>{%customdata[3]}</span><br><br>" +
    "<span style='color:#7F8C8D'>Total Spending:</span> <b style='color:#00B8A9'>{%customdata[4]}</b><br>" +
    "<span style='color:#7F8C8D'>Total Count:</span> <b style='color:#00B8A9'>{%customdata[5]:,.0f}</b><br>" +
    "<span style='color:#7F8C8D'>Avg Spending per Launch:</span> <b style='color:#00B8A9'>{%customdata[6]}</b>" +
    "<extra></extra>"
)

fig_px.update_traces(hovertemplate=hover_html, selector=dict(mode='markers'))

fig_px.update_yaxes(
    type='log',
    tickvals=[0.001,0.005, 0.01, 0.1, 1, 5],
    ticktext=['$1M', '$5M', '$10M', '$100M', '$1B', '$5B'],
    title_text='Average Cost per Launch by Program Over Time',
    title_font=dict(size=14, color='#F6416C', weight='bold'),
    tickfont=dict(size=13, color='#8898AA'),
)
```

```

        showgrid=True, gridcolor='#F0F2F6', zeroline=False
    )

    # We are going extract the traces (scatter and trendline) and design them.
    scatter_trace = fig_px.data[0]
    trendline_trace = fig_px.data[1]

    # scatter points
    scatter_trace.marker.size = 16
    scatter_trace.marker.color = '#00B8A9'
    scatter_trace.marker.line.width = 0.8
    scatter_trace.marker.line.color = 'DarkSlateGrey'

    # Trendlines
    trendline_trace.line.color = '#F6416C'
    trendline_trace.line.dash = 'dash'
    trendline_trace.line.width = 4

    # we are going to create shadows effect for the trendlines
    shadow_trace = go.Scatter(
        x=trendline_trace.x,
        y=trendline_trace.y,
        mode='lines',
        line=dict(color='rgba(220, 53, 69, 0.25)', width=15),
        hoverinfo='skip',
        showlegend=False
    )

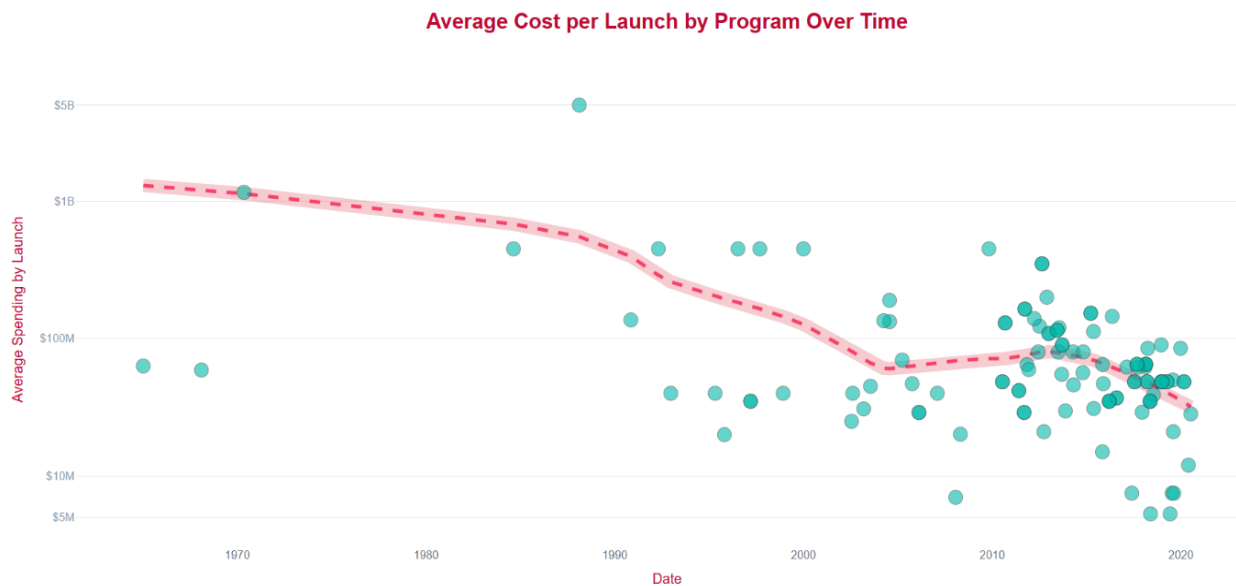
    fig = go.Figure( data=[shadow_trace, trendline_trace, scatter_trace],
        layout=fig_px.layout)

    fig.update_layout(
        font_family="Inter, Helvetica, Arial, sans-serif",
        title_font_size=24, title_font_color="#BF092F", title_x=0.5,
        xaxis_title='Date',
        yaxis_title='Average Spending by Launch',
        font=dict(family='Helvetica Neue, Arial, sans-serif', size=13,
        color='#5F6B78'),
        xaxis=dict(showgrid=False, zeroline=False, title_font_size=16,
        title_font_color="#BF092F"),
        yaxis=dict(showgrid=True, gridcolor='#E5E7EB', zeroline=False,
        title_font_size=16, title_font_color="#BF092F", type='log'),
        plot_bgcolor='rgba(0,0,0,0)',
        hoverlabel=dict(bgcolor="white", font_size=14, font_family="Inter, Helvetica,
        Arial, sans-serif", bordercolor="#E2E8F0")
    )

    fig.show()

```

Graph:



Interactive Graph Link: [Average Cost per Launch by Program Over Time](#)

The scatter plot and the LOWESS trendline clearly show a long-term decline in the average cost of rocket launches, particularly since the 1990s, suggesting that advances in aerospace technology, commercial competition, technologies for reusable launch vehicles, and standardized manufacturing processes have significantly improved the cost-effectiveness of access to space.

Another notable observation is the presence of several extreme outliers, such as Energiya, Saturn V, and the Space Shuttle orbiters, which reflect phases of the space race in which geopolitical ambitions and technological experimentation took precedence over financial sustainability.

In contrast, modern launch systems such as Falcon 1, Electron, and China's Long March series appear to operate at significantly lower costs, suggesting that contemporary aerospace programs are increasingly focusing on scalable, economically viable, and more cost-effective launch operations rather than prestige-driven megaprojects.

Analysis of Yearly Space Launch Activity and Mission Success

In this section, we are going to explore the evolution of global space launches from 1957 to 2020 based on the total number of launches per year.

This analysis illustrates how geopolitical events, technological breakthroughs, mass-market space exploration, and international competition have influenced the intensity of space exploration over time. In addition to the total number of launches, we also examine annual mission success rates to better understand how reliability and technical performance have improved throughout the history of modern rocket technology.

Extracting Year Information from Launch Dates

```
launch_year = df.copy()

# Extract the year, ensuring any NaT values are handled appropriately
launch_year['Year'] = df['Date'].dt.year.astype(int)
```

Aggregating Annual Launch Activity

The dataset is grouped by year using pandas' aggregation functions to calculate the total number of launches conducted annually across all countries and organizations.

```
# Total Launch Count By year
total_launch_count_by_year = launch_year.groupby('Year', as_index=False)
                                .agg(Total_Launch_Count=("Mission_Status", "count"))
```

Filtering Successful Missions and Combining Annual Statistics

The dataset is filtered using the `.query()` function to select only successful missions. The successful launches are then grouped by year to calculate the success rates.

The resulting dataset is then merged with the general annual launch statistics, allowing the total number of launches, successful launches, failed launches, and success rates to be analyzed together in a single chronological DataFrame.


```
# Total Successful Launch Count by year
successful_launches_by_year = launch_year.query("Mission_Status == 'Success'")
successful_launches_by_year = successful_launches_by_year.groupby(["Year"],
as_index=False).agg(Total_Successful_Launch_Count=("Mission_Status", "count"))

# Merging the two dataframes:
launch_by_year = total_launch_count_by_year.merge(successful_launches_by_year,
on="Year", how="left")

launch_by_year.fillna(0, inplace=True)
```

Calculating Annual Success Rates and Failure Counts

Based on the total number of launches and the number of successful launches, a new success rate metric is calculated to measure the annual reliability of the missions as a percentage.

The data is then sorted chronologically and supplemented with a parameter for the number of failed launches, allowing both successful and failed missions to be visualized simultaneously over time.

```
# Success Rate:
launch_by_year["Success_Rate"] =
    launch_by_year.Total_Successful_Launch_Count / launch_by_year.Total_Launch_Count * 100

launch_by_year.sort_values('Year', ascending=True, inplace=True)

# Prepare the Data: Calculate the failed launches
launch_by_year['Failed_Launch_Count'] =
    launch_by_year['Total_Launch_Count'] - launch_by_year['Total_Successful_Launch_Count']
```

Creating the Stacked Bar Chart Visualization

Using Plotly chart objects, a stacked bar chart is created to display successful and failed launches for each year in a single visualization.

This approach makes it easier to compare launch frequency and mission reliability simultaneously, while interactive hover data provides additional annual statistics—such as the total number of launches and success rates—for a more in-depth analysis.

code:

```
fig = go.Figure()

fig.add_trace(go.Bar(
    x=launch_by_year['Year'],
    y=launch_by_year['Total_Successful_Launch_Count'],
    name='Successful',
    marker_color='#3F9AAE',
    marker_line_width=0,      # Removes borders for a cleaner, flat-design look
    customdata=launch_by_year[['Total_Launch_Count', 'Success_Rate',
'Failed_Launch_Count']],
    hovertemplate=(
        "<b style='font-size:16px; color:#1A252F'>{x}</b><br><br>" +
        "<span style='color:#7F8C8D'>Successful Launches:</span> <b style='color:#3F9AAE'>{y}</b><br>" +
        "<span style='color:#7F8C8D'>Failed Launches:</span> <b style='color:#F96E5B'>{customdata[2]}</b><br><br>" +
        "<span style='color:#7F8C8D'>Total Launches:</span> <b style='color:#3F9AAE'>{customdata[0]}</b><br>" +
        "<span style='color:#7F8C8D'>Success Rate:</span> <b style='color:#3F9AAE'>{customdata[1]:.1f}%</b>" +
        "<extra></extra>"
    )
))

fig.add_trace(go.Bar(
    x=launch_by_year['Year'],
    y=launch_by_year['Failed_Launch_Count'],
    name='Failed',
    marker_color='#F96E5B',
    marker_line_width=0,
    customdata=launch_by_year[
        ['Total_Launch_Count', 'Success_Rate', 'Total_Successful_Launch_Count']],
    hovertemplate=(
        "<b style='font-size:16px; color:#1A252F'>{x}</b><br><br>" +
        "<span style='color:#7F8C8D'>Failed Launches:</span> <b style='color:#F96E5B'>{y}</b><br>" +
        "<span style='color:#7F8C8D'>Successful Launches:</span> <b style='color:#3F9AAE'>{customdata[2]}</b><br><br>" +
        "<span style='color:#7F8C8D'>Total Launches:</span> <b style='color:#3F9AAE'>{customdata[0]}</b><br>" +
        "<span style='color:#7F8C8D'>Success Rate:</span> <b style='color:#3F9AAE'>{customdata[1]:.1f}%</b>" +
        "<extra></extra>"
    )
))

fig.update_layout(
    barmode='stack', # we use stack chart here
    template='plotly_white',
    title=dict(
        text='<b style="color:#134686">Space Mission Launches Over Time</b><br>'
        '<span style="font-size:14px; color:#134686">Annual volume of'
        'successful vs. failed orbital attempts</span>',
        font=dict(size=22, family='Arial, sans-serif', color='#1A252F'),
        x=0.02,
        xanchor='left'
    ),
)
```

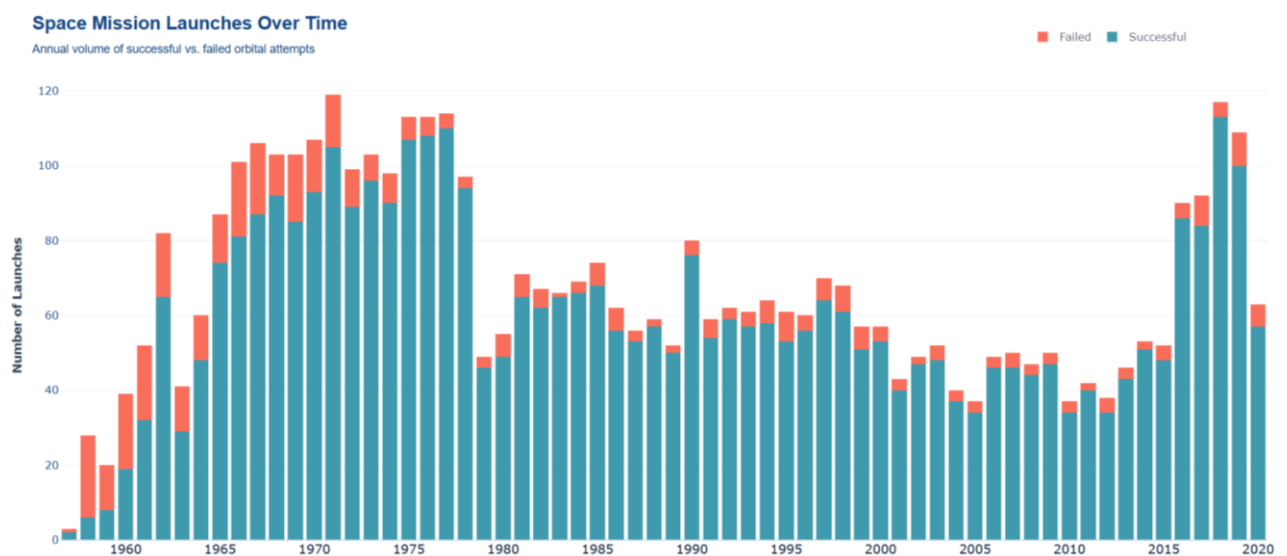
```

legend=dict(orientation="h", x=0.8, y=1.1, title_text='', font=dict(size=13),
xaxis=dict(
    title='',
    showgrid=False,
    tickfont=dict(size=15, color='#132440'),
    tickmode='linear',
    dtick=5, # ticks for every five year
),
yaxis=dict(
    title='Number of Launches',
    title_font=dict(size=14, color='#2C3E50', weight='bold'),
    tickfont=dict(size=13, color='#134686'),
    showgrid=True, gridcolor='#F0F2F6',
    zeroline=True, zerolinecolor='#E2E8F0', zerolinewidth=2
),
margin=dict(l=20, r=20, t=100, b=20),
hoverlabel=dict(bgcolor="white", font_size=13, bordercolor="#E2E8F0"),
plot_bgcolor='rgba(0,0,0,0)',
paper_bgcolor='rgba(0,0,0,0)'
)

# fig.show()

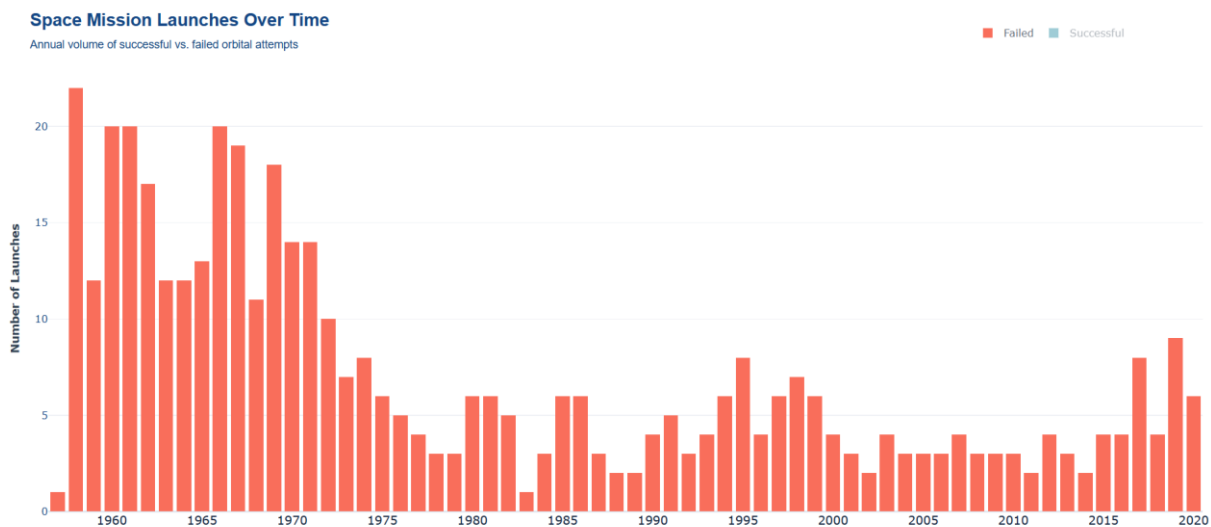
```

Graph:



Interactive Graph Link: [Space Mission Launches Over Time](#)

Filtered for Failures:



The graphs show that while rocket technology and mission reliability improved dramatically over time, the total number of launches declined after the peak years of the Cold War and remained relatively stable between 1979 and 2016, at around 60 to 80 launches per year; This decline was largely due to the end of the intense space race between the U.S. and the USSR, diminishing military urgency, economic crises in the Soviet Union in the late 1970-80s, and the gradual shift from a competition to one involving fewer, but more advanced and capable satellites.

The sharp decline around 1978–1979 is closely linked to the slowdown in early Soviet heavy-lift launch programs, shifting geopolitical priorities, and the maturation of space technology, as newer rockets became more reliable and satellites far more efficient.

Starting in 2016, launch activity began to increase rapidly again, primarily due to the rise of the commercial space industry—particularly companies like SpaceX—as well as the massive expansion of China’s national space program.

At the same time, the second graph illustrates one of the most significant achievements of modern aerospace technology: Despite the significantly greater complexity of launches today, the number of annual launch failures has dropped dramatically compared to the early days of the space age, demonstrating how decades of technical standardization, reusable launch systems, improved computing power, and international experience have significantly increased the reliability of launches.

Temporal Analysis of Monthly Space Launch Activity

In this section, we are going to analyze how satellite launches are distributed across the different months of the year to determine whether global space missions follow seasonal or operational patterns.

Analyzing the monthly launch frequency helps reveal how factors such as weather conditions, orbital mechanics, the climate at the launch site, budget cycles, and mission planning can influence the timing of space missions. With this analysis, we aim to determine whether certain months consistently see higher launch activity and assess whether the global aerospace industry operates according to observable temporal trends throughout the calendar year.

Temporal Feature Engineering and Monthly Launch Aggregation

To perform this analysis, we first expanded the dataset by using Pandas' date and time functions to extract additional time-related features from the original "Date" column, specifically the year and month. After generating these categorical time-based features, we grouped the dataset by month and aggregated the number of launches. Finally, the resulting DataFrame was sorted by the total number of launches.

```
launch_month = df.copy()

# month and year column for every entry
df['Year'] = df['Date'].dt.year.astype(int)
df['Month'] = df['Date'].dt.month_name()

# Grouping by Months and adding the number of Launches to the next
launches_per_month = df.groupby('Month',
as_index=False).agg(Total_Launch_Count=("Mission_Status", "count"))

# Sort the months by total Launches
launches_per_month = launches_per_month.sort_values('Total_Launch_Count',
ascending=False)
```

Creating Bar Chart for Monthly Launch Distribution

To illustrate the distribution of launches across the months, we used a Plotly bar chart in which each bar represents the total number of launches in a given month.

code:

```
m_bar = px.bar(
    launches_per_month,
    x='Month',
    y='Total_Launch_Count',
    text='Total_Launch_Count', # Adds the exact number directly above the bar
    color_continuous_scale='Bluy1',
    color="Total_Launch_Count"
)

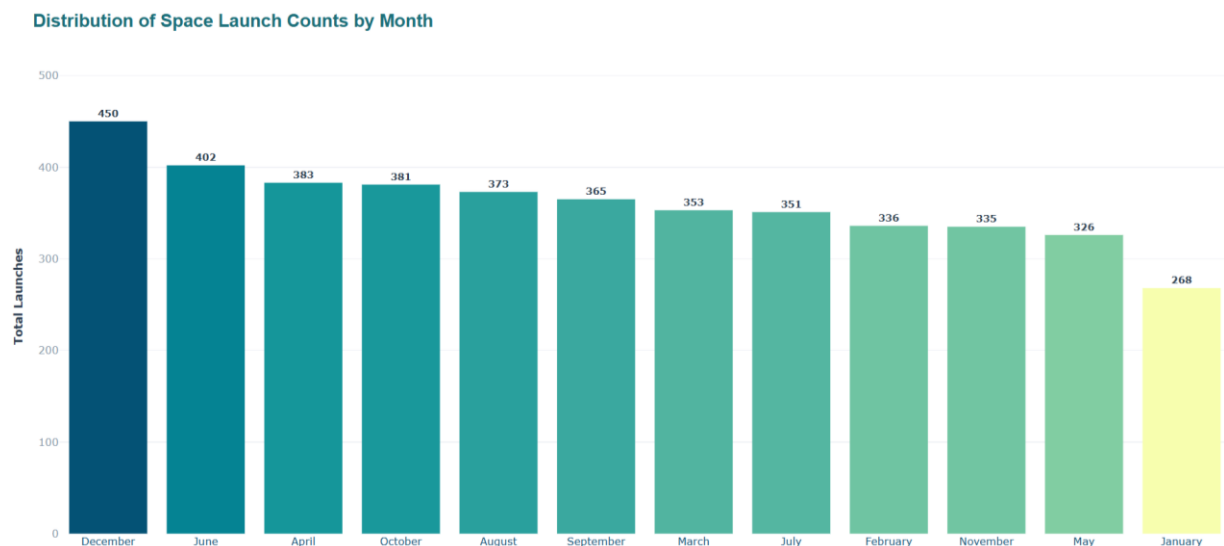
# Refining the bars and texts
m_bar.update_traces(
    marker_line_width=0, # Removes borders for a flat, modern design
    texttemplate='<b>{%text}</b>',
    textposition='outside', # Places the numbers cleanly above the bars
    textfont=dict(size=12, color='#2C3E50'),
    # Custom Hover Template
    hovertemplate=(
        "<b style='font-size:15px; color:#2C3E50'>{%x}</b><br><br>" +
        "<span style='color:#467289'>Total Launches:</span> +
        "<b style='color:#0F4C75'>{%y}</b>" +
        "<extra></extra>"
    )
)

# Layout and Typography Arrangements:
m_bar.update_layout(
    template='plotly_white',
    title=dict(
        text='<b>Distribution of Space Launch Counts by Month</b>',
        font=dict(size=22, family='Arial, sans-serif', color='#1B6F7A'),
        x=0.02,
        xanchor='left'
    ),
    xaxis=dict(
        title='',
        tickfont=dict(size=13, color='#0F4C75'),
        showgrid=False,
        categoryorder='total descending'
    ),
    yaxis=dict(
        title='Total Launches',
        title_font=dict(size=14, color='#2C3E50', weight='bold'),
        tickfont=dict(size=13, color='#8898AA'),
        showgrid=True, gridcolor='#F0F2F6', # soft grid lines
        zeroline=False,
        range=[0, launches_per_month['Total_Launch_Count'].max() * 1.15]),
    margin=dict(l=20, r=20, t=80, b=20),
```

```

        hoverlabel=dict(bgcolor="white", font_size=13, bordercolor="#E2E8F0"),
        plot_bgcolor='rgba(0,0,0,0)', paper_bgcolor='rgba(0,0,0,0)',
        coloraxis_showscale=False,
    )
m_bar.show()

```



Graph Link: [Distribution of Space Launch Counts by Month](#)

The monthly distribution of launches does not reveal any particularly pronounced seasonal pattern, as launch activity is spread relatively evenly throughout the year, with December clearly standing out as the month with the highest number of launches, while January sees the lowest level of activity.

This pattern is likely influenced more by organizational factors than by astronomy itself; many companies strive to complete missions before the end of the fiscal or reporting year, while the operational pace often slows in January—due to schedule delays following the holidays, weather-related restrictions at major launch sites, and the preparation phase for new annual mission cycles.

From a statistical perspective, the differences between most months are not large enough to confidently speak of a strict seasonal launch pattern, meaning the results are only moderately significant.

Statistical Analysis of Annual Space Mission Expenditures

In this section, we are going to examine how the financial scale of space missions has evolved over time by analyzing annual launch costs based on missions.

In contrast to the previous program-specific analysis, this approach focuses on the general annual spending patterns of the global space industry, allowing us to observe long-term economic trends, fluctuations, and periods of unusually high investment in space exploration.

Using statistical aggregation and trend analysis, we aim to understand how the economics of launches have changed across different technological eras and how factors such as the space race, privatization, reusable rockets, and modern satellite deployment strategies have influenced mission costs over time.

Annual Aggregation and Statistical Feature Extraction

To analyze annual launch costs, firstly, the dataset is filtered to include only launches for which price data was available and then the launches are grouped by year.

As part of this grouping process, we calculated several key statistical metrics, including minimum, maximum, and average values, the median, percentile ranges, standard deviation, total annual expenditures, and the number of launches with available price data. This allowed us to assess not only the average costs but also the dispersion, volatility, and distribution characteristics of launch costs across the various years.

```
launch_prices = df.dropna(subset=['Price'])

# Group by Year and calculate statistics
price_trends = launch_prices.groupby("Year", as_index=False).agg(
    Min_Price=("Price", "min"),
    Max_Price=("Price", "max"),
    Avg_Price=("Price", "mean"),
    pctl_25=("Price", ("q25", lambda x: x.quantile(0.25))),
    Median=("Price", "median"),
    pctl_75=("Price", ("q75", lambda x: x.quantile(0.75))),
    Standard_Deviation=("Price", "std"),
    Total_Cost=("Price", "sum"),
    Launch_Count_Price_Available=("Price", "count"))
```


Integrating Complete Launch Activity into the Financial Dataset

Total number of launches per year is additionally calculated without removing missing values and these entries are incorporated into the financial data framework.

```
# creating a new grouped dataframe and merge with main dataframe
price_available = df.groupby("Year",
as_index=False).agg(Total_Launch_Count=("Mission_Status", "count"))
price_trends = price_trends.merge(price_available, on='Year', how='left')
```

Applying Financial Display Formatting

Previously defined formatting functions are reused to convert large numerical output values into more readable financial figures expressed in millions and billions of dollars.

```
# displaying-format for the price values:
price_trends['Total_Cost_Fmt'] = price_trends['Total_Cost'].apply(format_spending_label)
price_trends['Avg_Price_Fmt'] = price_trends['Avg_Price'].apply(format_spending_label)
price_trends['Min_Price_Fmt'] = price_trends['Min_Price'].apply(format_spending_label)
price_trends['Max_Price_Fmt'] = price_trends['Max_Price'].apply(format_spending_label)
price_trends['Median_Fmt'] = price_trends['Median'].apply(format_spending_label)

# Price layout for the y-axis
price_trends['Total_Cost_B'] = price_trends['Total_Cost'] / 1000
```

Smoothing Long-Term Trends with Moving Averages

To better highlight long-term spending trends and smooth out short-term annual fluctuations, we have calculated a three-year moving average.

This smoothing method helps reveal broader financial trends in the aerospace industry by minimizing the visual impact of extreme outliers in mission costs and irregular annual spending spikes.

```
# year moving average so that lines has a tune
moving_average = price_trends['Total_Cost_B'].rolling(window=3).mean()
```

Creating Trendline-Scatter Graph for Annual Launch Expenditures

We use an interactive scatter plot combined with an overlapped trend line to illustrate the trend in total annual spending on space launches across different eras of space exploration, while ensuring that individual annual data points and statistical fluctuations remain visible.

code:

```
fig = go.Figure()

# Scatter graph
fig.add_trace(go.Scatter(
    x=price_trends['Year'],
    y=price_trends['Total_Cost_B'],
    mode='markers',
    name='Total Expenditure',
    marker=dict(color='#467289', size=15, opacity=0.8, line=dict(width=1.5,
color='white')),
    # for hovertemplate:
    customdata=price_trends[[
        'Total_Cost_Fmt', 'Avg_Price_Fmt', 'Min_Price_Fmt',
        'Max_Price_Fmt', 'Median_Fmt', 'Launch_Count_Price_Avaliable',
'Total_Launch_Count'
    ]],
    hovertemplate=(
        "<b style='font-size:16px; color:#1A252F'>Year: {x}</b><br><br>" +
        "<span style='font-size:13px; color:#7F8C8D'>Total Spending:</span>" +
        "<b style='font-size:13px; color:#467289'>{customdata[0]}</b><br><br>" +
        "<span style='color:#7F8C8D'>Average Price:</span> <b style='color:#467289'>{customdata[1]}</b><br>" +
        "<span style='color:#7F8C8D'>Min Price:</span> <b style='color:#467289'>{customdata[2]}</b><br>" +
        "<span style='color:#7F8C8D'>Max Price:</span> <b style='color:#467289'>{customdata[3]}</b><br>" +
        "<span style='color:#7F8C8D'>Median Price:</span> <b style='color:#467289'>{customdata[4]}</b><br><br>" +
        "<span style='color:#7F8C8D'>Launches (Cost Known):</span>" +
        "<b style='color:#467289'>{customdata[5]}</b><br>" +
        "<span style='color:#7F8C8D'>Total Launches:</span> <b style='color:#467289'>{customdata[6]}</b>" +
        "<extra></extra>"
    )
))

# Rolling Average Line
fig.add_trace(go.Scatter(
    x=price_trends['Year'],
    y=moving_average,
    mode='lines',
    name='3-Year Rolling Average',
    line=dict(color='#D9534F', width=3, shape='spline'),
    hoverinfo='skip'
))

# Layout Styling
```

```

fig.update_layout(
    template='plotly_white',
    title=dict(
        text='<b>Total Expenditure on Satellite Launches by Year</b>',
        font=dict(size=22, family="Helvetica Neue, Arial, sans-serif",
color="#1A252F"),
        y=0.95, x=0.02,
        xanchor='left', yanchor='top'
    ),
    legend=dict(
        orientation="h",
        yanchor="bottom", xanchor="right",
        y=1.02, x=0.98,
        bgcolor="rgba(255, 255, 255, 0)",
        font=dict(size=13, color='#5F6B78')
    ),
    hovermode='closest', # this renders custom HTML cards
    hoverlabel=dict(bgcolor="white", font_size=13, bordercolor="#E2E8F0"),
    margin=dict(l=60, r=40, t=100, b=60),
    plot_bgcolor='rgba(0,0,0,0)',
    paper_bgcolor='rgba(0,0,0,0)'
)

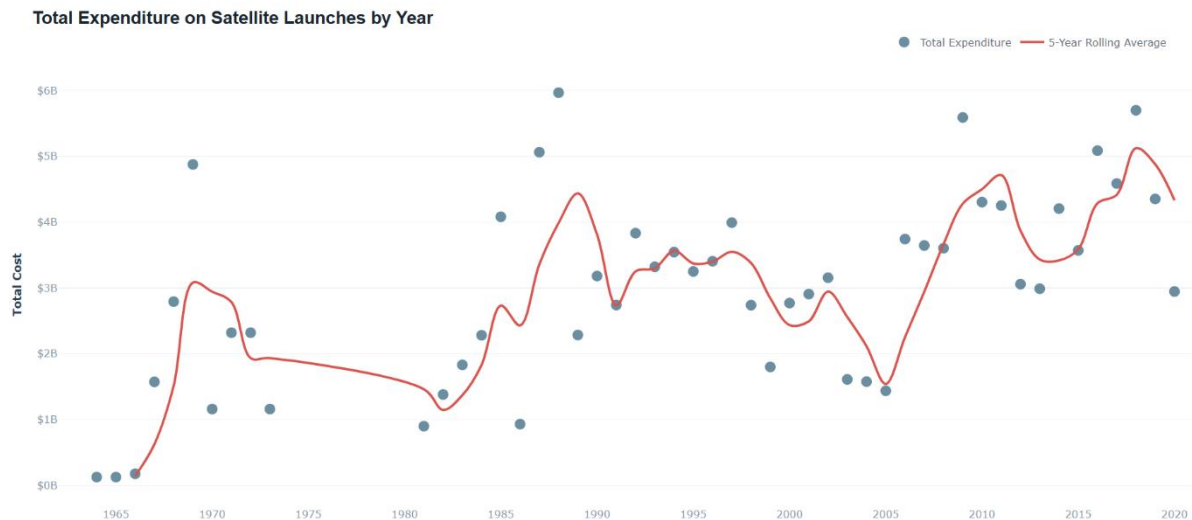
# Axes Arrangements
fig.update_xaxes(
    tickangle=0,
    tickmode='linear',
    tick0=1965, dtick=5,
    showgrid=False,
    tickfont=dict(size=13, color='#8898AA'),
    range=[1962, 2021]
)

fig.update_yaxes(
    title=dict(text="Total Cost", font=dict(size=14, color='#2C3E50',
weight='bold')),
    showgrid=True,
    gridcolor='#F0F2F6', # soft grid lines
    tickformat='$.0f', # Formats numbers as $2, $4, $6
    ticksuffix='B', # Appends a 'B' to make it $2B, $4B
    tickfont=dict(size=13, color='#8898AA'),
    zeroline=False
)

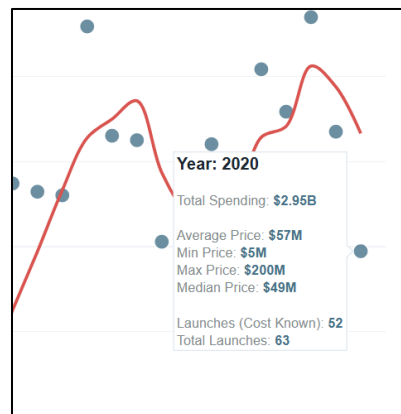
# fig.write_html("interactive_graphs/space_race/total_cost_by_year.html")

fig.show()

```



Two points from the Graph:



Interactive Graph Link: [Total Expenditure on Satellite Launches by Year](#)

The graph illustrates a profound long-term shift in the cost-effectiveness of spaceflight: While total annual spending has remained relatively high over the decades, the average cost per launch has fallen significantly since the 2000s, particularly in the late 2010s, when the frequency of launches increased rapidly.

Earlier periods, particularly the Apollo and Space Shuttle eras, were characterized by extremely expensive, government-funded missions, in which average annual launch costs often exceeded several hundred million dollars, while modern programs have increasingly shifted to more cost-effective launch systems.

One particularly important observation is that, despite significantly lower average launch prices, total spending remained very high in years such as 2018, suggesting that the modern space industry has achieved greater launch capacity and operational efficiency rather than simply scaling back its activities.

The decline in both median and average costs strongly suggests that reusable rocket technology, competition from the private sector and the rise of private satellite constellations have fundamentally transformed the financial structure of global access to space, making launches more economically sustainable and scalable compared to earlier space programs.

Evolution of Launch Activity Among Major Space Organizations

in this section, we are going to explore how the annual launch activities of the world's leading space agencies have evolved over time and how the balance of power among the various players has shifted throughout the history of space exploration.

By comparing launch frequencies year over year, we can better understand the rise and fall of major government and private organizations and identify key historical turning points such as the Space Race, post-Cold War restructuring, and the emergence of private space companies.

This analysis also illustrates how technological innovations, geopolitical competition, and commercialization have transformed the global space industry from a government-dominated sector into a far more competitive and diversified ecosystem.

Aggregating and Filtering Launch Activity Data

To prepare the dataset for time-series analysis, the dataset is first grouped the race data by both year and organization. Then the ten organizations with the highest cumulative number of launch are identified across the dataset. This allowed us to reduce visual clutter while retaining the historically most influential organizations.

Finally, we integrated country information into the grouped data frames so that each organization could later be interpreted within its geopolitical and industrial context during the visualization phase.

```

top10_org_counts = (df.groupby(['Year', 'Organisation'])
                    .agg(Total_Launch_Count=("Mission_Status", "count"))
                    .reset_index())

# Get the top 10 organizations with the most total launches
top_10_orgs = top10_org_counts.groupby('Organisation')['Total_Launch_Count']
                    .sum()
                    .nlargest(10).index

top10_org_counts =
    top10_org_counts[top10_org_counts['Organisation'].isin(top_10_orgs)]

df_country_organisation = df.groupby(["Organisation"])
                    .agg(Country=("Country", "max"))

top10_org_counts = top10_org_counts.merge(
    df_country_organisation, on='Organisation', how='inner')

```

Creating Multi-Line Time Series Chart

We use a multi-line Plotly visualization to compare trends in the launch activities of various organizations over time. This allows us to observe leadership changes, long-term growth patterns, and periods of decline within the global space industry.

code:

```

# high-contrast color palette
color_palette = [
    '#1f77b4', '#d62728', '#2ca02c', '#ff7f0e', '#9467bd',
    '#8c564b', '#e377c2', '#7f7f7f', '#bcbd22', '#17becf'
]

fig = px.line(
    top10_org_counts,
    x='Year',
    y='Total_Launch_Count',
    color='Organisation',
    custom_data=['Organisation', 'Country'],
    color_discrete_sequence=color_palette
)

# html cover cards
hover_html = (
    "<b style='font-size:16px; color:#1A252F'>{%customdata[0]}</b><br>" +
    "<span style='font-size:13px; color:#5F6B78'>{%customdata[1]}</span><br><br>" +

```

```

" <span style='color:#7F8C8D'>Year:</span> <b style='color:#2C687B'>{%x}</b><br>" +
" <span style='color:#7F8C8D'>Launches:</span> <b style='color:#2C687B'>{%y}</b>" +
"<extra></extra>"
)

fig.update_traces(
    mode='lines',          # This removes the noisy dots
    line=dict(width=2.5),
    hovertemplate=hover_html
)

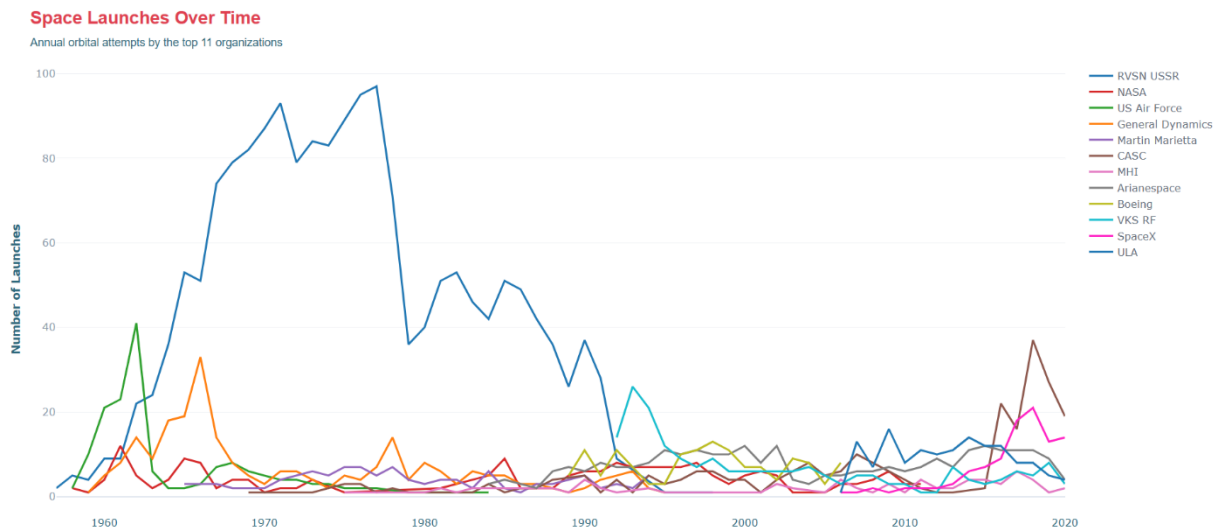
fig.update_layout(
    template='plotly_white',
    title=dict(
        text='<b>Space Launches Over Time</b><br>
        <span style="font-size:14px; color:#2C687B">
            Annual orbital attempts by the top 10 organizations</span>',
        font=dict(size=22, family='Arial, sans-serif', color='#E03F4F'),
        x=0.02,
        xanchor='left'
    ),
    xaxis=dict(
        title='',
        showgrid=False,
        tickfont=dict(size=13, color='#2C687B'),
        tickmode='linear',
        dtick=10 # Shows every 10 years to keep the x-axis clean
    ),
    yaxis=dict(
        title='Number of Launches',
        title_font=dict(size=14, color='#2C687B', weight='bold'),
        tickfont=dict(size=13, color='#8898AA'),
        showgrid=True,
        gridcolor='#F0F2F6',
        zeroline=True,
        zerolinecolor='#E2E8F0',
        zerolinewidth=2,
    ),
    legend=dict(title_text='', font=dict(size=13, color='#5F6B78')),
    margin=dict(l=20, r=20, t=100, b=20),
    hoverlabel=dict(bgcolor="white", font_size=15, bordercolor="#E2E8F0"),
    plot_bgcolor='rgba(0,0,0,0)',
    paper_bgcolor='rgba(0,0,0,0)'
)

#
fig.write_html("interactive_graphs/space_race/top_10_organizations_by_year.html")

fig.show()

```

Graph:



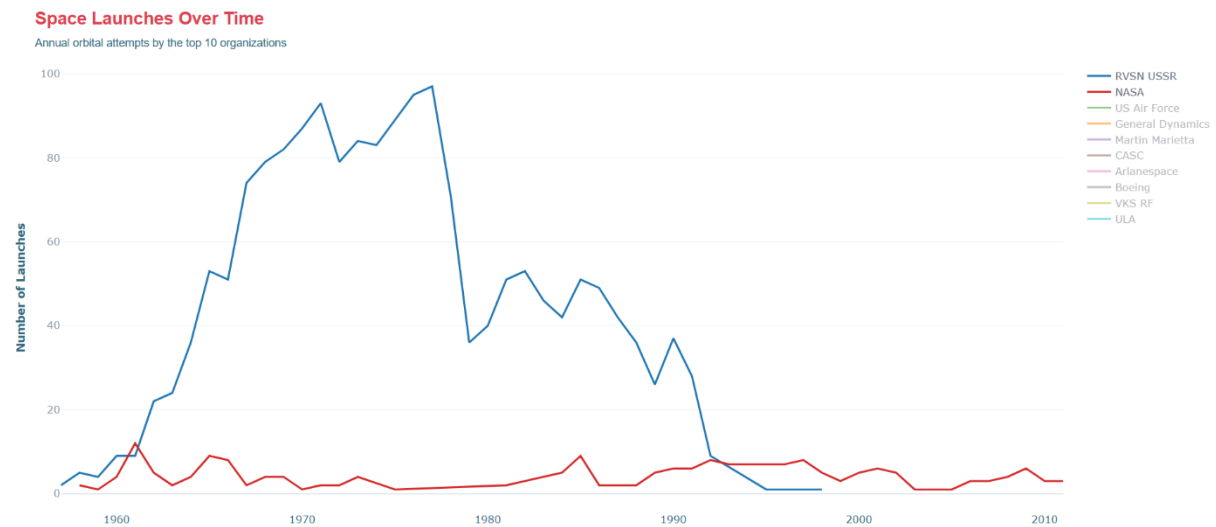
Interactive Graph Link: [Space Launches Over Time](#)

The chart clearly illustrates how leadership in space launches has evolved over time: from the strong dominance of Soviet military organizations such as the USSR's Strategic Rocket Forces (RVSN) during the Cold War to a far more diverse and leveraged ecosystem.

In the 1960s and 1970s, launch activities were heavily concentrated among a few government-funded organizations, driven by geopolitical rivalries, while after the 2000s, the market increasingly shifted toward private providers such as SpaceX, ULA, and international organizations like CASC and Arianespace.

A particularly important insight is that the modern space industry is no longer dependent on a single dominant institution; instead, competition, privatization, and international investment have created a more sustainable and scalable launch environment with significantly greater operational flexibility.

NASA vs RVSN USSR:



The comparison between NASA and the Soviet Union's Strategic Rocket Forces clearly reflects the dynamics of the space race during the Cold War, in which the Soviet Union achieved an extraordinarily high launch frequency in the 1960s and 1970s due to the aggressive deployment of military satellites, the development of intercontinental ballistic missiles, and centralized state investment.

However, the dramatic decline in the activities of the USSR's Strategic Rocket Forces after the late 1980s is closely linked to the economic decline and eventual dissolution of the Soviet Union in 1991, while NASA maintained a smaller but more stable long-term launch presence, focusing on scientific exploration, human spaceflight, and technological research rather than concentrating exclusively on the number of launches.

Space Launches Over Time

Annual orbital attempts by the top 10 organizations

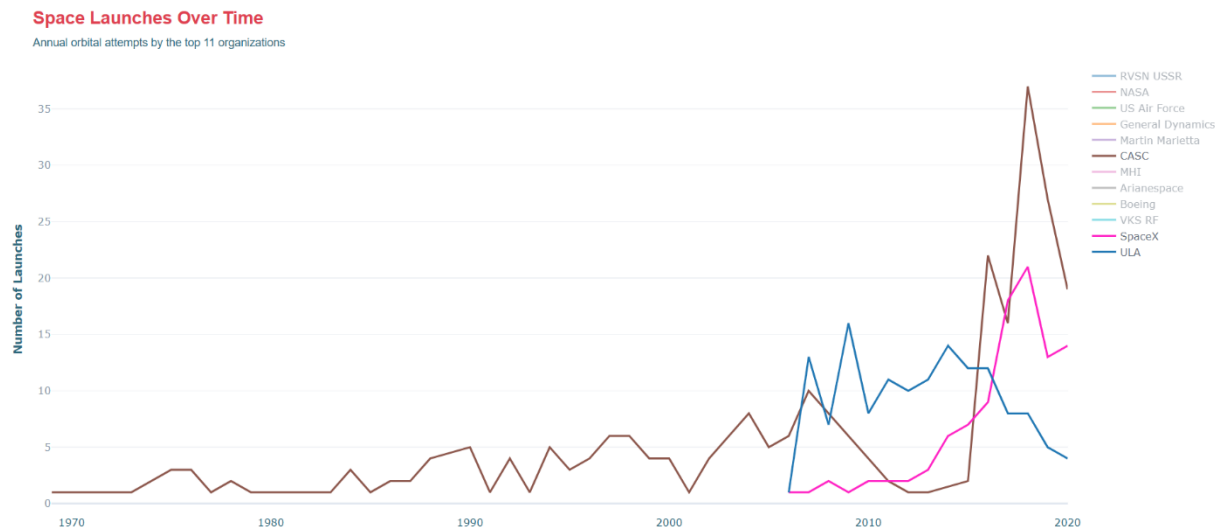
The chart displays the annual number of orbital launches by the top 10 organizations from 1960 to 2020. The Y-axis represents the 'Number of Launches' (0 to 35), and the X-axis represents the 'Year' (1960 to 2020). The data shows a long period of low launch activity (mostly 1-5 launches per year) until the mid-2010s, followed by a sharp increase. The Russian Space Forces (RVSN USSR) and NASA are the most active organizations, with RVSN USSR showing a massive spike in 2018.

Year	RVSN USSR	NASA	US Air Force	General Dynamics	Martin Marietta	CASC	Arianespace	Boeing	VKS RF	ULA
1960	0	2	0	0	0	0	0	0	0	0
1961	0	4	0	0	0	0	0	0	0	0
1962	0	12	0	0	0	0	0	0	0	0
1963	0	5	0	0	0	0	0	0	0	0
1964	0	2	0	0	0	0	0	0	0	0
1965	0	4	0	0	0	0	0	0	0	0
1966	0	9	0	0	0	0	0	0	0	0
1967	0	8	0	0	0	0	0	0	0	0
1968	0	2	0	0	0	0	0	0	0	0
1969	0	4	0	0	0	0	0	0	0	0
1970	0	4	0	0	0	1	0	0	0	0
1971	0	1	0	0	0	1	0	0	0	0
1972	0	2	0	0	0	1	0	0	0	0
1973	0	4	0	0	0	1	0	0	0	0
1974	0	1	0	0	0	3	0	0	0	0
1975	0	1	0	0	0	3	0	0	0	0
1976	0	1	0	0	0	1	0	0	0	0
1977	0	1	0	0	0	3	0	0	0	0
1978	0	1	0	0	0	1	0	0	0	0
1979	0	2	0	0	0	2	0	0	0	0
1980	0	2	0	0	0	2	0	0	0	0
1981	0	2	0	0	0	2	0	0	0	0
1982	0	3	0	0	0	2	0	0	0	0
1983	0	5	0	0	0	2	0	0	0	0
1984	0	9	0	0	0	2	0	0	0	0
1985	0	1	0	0	0	3	0	0	0	0
1986	0	2	0	0	0	2	0	0	0	0
1987	0	2	0	0	0	2	0	0	0	0
1988	0	2	0	0	0	4	0	0	0	0
1989	0	5	0	0	0	5	0	0	0	0
1990	0	6	0	0	0	1	0	0	0	0
1991	0	6	0	0	0	1	0	0	0	0
1992	0	8	0	0	0	1	0	0	0	0
1993	0	7	0	0	0	1	0	0	0	0
1994	0	7	0	0	0	5	0	0	0	0
1995	0	7	0	0	0	3	0	0	0	0
1996	0	7	0	0	0	4	0	0	0	0
1997	0	7	0	0	0	6	0	0	0	0
1998	0	8	0	0	0	6	0	0	0	0
1999	0	3	0	0	0	4	0	0	0	0
2000	0	4	0	0	0	1	0	0	0	0
2001	0	6	0	0	0	1	0	0	0	0
2002	0	1	0	0	0	8	0	0	0	0
2003	0	1	0	0	0	5	0	0	0	0
2004	0	1	0	0	0	8	0	0	0	0
2005	0	1	0	0	0	5	0	0	0	0
2006	0	3	0	0	0	10	0	0	0	0
2007	0	4	0	0	0	7	0	0	0	0
2008	0	5	0	0	0	4	0	0	0	0
2009	0	3	0	0	0	1	0	0	0	0
2010	0	3	0	0	0	1	0	0	0	0
2011	0	1	0	0	0	1	0	0		

In contrast, NASA's visible launch numbers have declined since 2011, not because the United States has scaled back its space activities, but because NASA fundamentally changed its operational strategy by retiring the Space Shuttle and outsourcing launch operations to private providers such as SpaceX and ULA.

This transition represents one of the most significant structural shifts in modern aerospace history: The United States has moved from a government-operated launch model to a public-private ecosystem in which NASA increasingly acts as a contracting authority, research coordinator, and program manager for space exploration, rather than as a direct operator of launch vehicles.

CASC vs SPACEX + ULA:



A comparison between CASC and the joint rise of SpaceX and ULA highlights two fundamentally different models of space development: China's centrally coordinated, state-directed expansion versus the increasingly privatized and competitive commercial ecosystem of the United States.

The steady growth of CASC demonstrates the effectiveness of long-term national planning and sustained government investment, yet the rapid rise of SpaceX—particularly since the mid-2010s—shows how reusable rocket technology, commercial competition, and collaboration with government agencies have drastically increased launch frequency while simultaneously lowering operating costs.

Although China's progress is undeniably significant, the chart also shows that the U.S.'s hybrid cooperation model between NASA and private aerospace companies has created an extremely adaptable system capable of scaling launch operations more quickly than traditional, purely government-led structures.

Cold War Space Race: USA vs Soviet Union

In this section, we are going to focus specifically on the space race between the United States and the Soviet Union during the Cold War, one of the most influential technological competitions in modern history.

By comparing the annual launch activities of the two superpowers, we aim to understand how geopolitical tensions, military competition, and scientific ambition directly shaped the development of global space exploration from the late 1950s until the dissolution of the Soviet Union in 1991.

This analysis is particularly valuable because the space race was not only a scientific competition but also a demonstration of industrial power, economic strength, and ideological superiority, making launch frequency a key indicator of the national strategic priorities of that era.

Aggregating Cold War Launch Activity

To prepare the dataset for historical comparisons, we grouped the launch data by both country and year, which allowed us to calculate the total number of launches each country conducted annually over the entire duration of the dataset.

This aggregation process transforms the raw mission-level data into a structured time-series dataset, making it possible to analyze long-term launch trends.

```
cold_war_data = df.groupby(['Country', 'Year'],  
as_index=False).agg(Total_Launches=('Mission_Status', 'count'))
```

Constructing the Soviet Union Dataset

Since, historically, many Soviet launches were conducted from launch sites in what is now Kazakhstan we have combined launch data from Kazakhstan and the Russian Federation to provide a more accurate picture of the operational activities of the former Soviet Union. After filtering these countries, we combined their annual launch counts into a single dataset for "Soviet Union".

Finally, we reintegrated this reconstructed Soviet dataset into the main data frame and filtered the results to include only the United States and the Soviet Union.

```

soviet_union_data = cold_war_data[cold_war_data["Country"]
                                   .isin(["Russian Federation", "Kazakhstan"])]

soviet_union_data = soviet_union_data.groupby("Year")["Total_Launches"].sum()

soviet_union_data["Country"] = "Soviet Union"

cold_war_data = pd.concat([cold_war_data, soviet_union_data], ignore_index=True)

usa_and_soviet = cold_war_data.loc[cold_war_data["Country"]
                                   .isin(["Soviet Union", "USA"])]

```

Creating the Comparative Time-Series Visualization

code:

```

thematic_colors = ['#1A5F7A', '#D9534F']

l_chart = px.line(
    usa_and_soviet,
    x='Year',
    y='Total_Launches',
    color='Country',
    color_discrete_sequence=thematic_colors
)

hover_html = (
    "<b style='font-size:16px; color:#1A252F'>{%data.name}</b><br><br>" +
    "<span style='color:#7F8C8D'>Year:</span> <b style='color:#7D5A50'>{%x}</b><br>" +
    "<span style='color:#7F8C8D'>Total Launches:</span> <b style='color:#7D5A50'>{%y}</b>" +
    "<extra></extra>"
)

l_chart.update_traces(line=dict(width=3), hovertemplate=hover_html)

l_chart.update_layout(
    template='plotly_white',
    title=dict(
        text='<b>The Space Race: USA vs. Soviet Union</b><br>'
        + '<span style="font-size:14px; color:#7F8C8D"> + '
        + 'Annual orbital launch attempts by superpower</span>',
        font=dict(size=22, family='Arial, sans-serif', color='#1A252F'),
        x=0.01, xanchor='left'
    ),
    xaxis=dict(

```

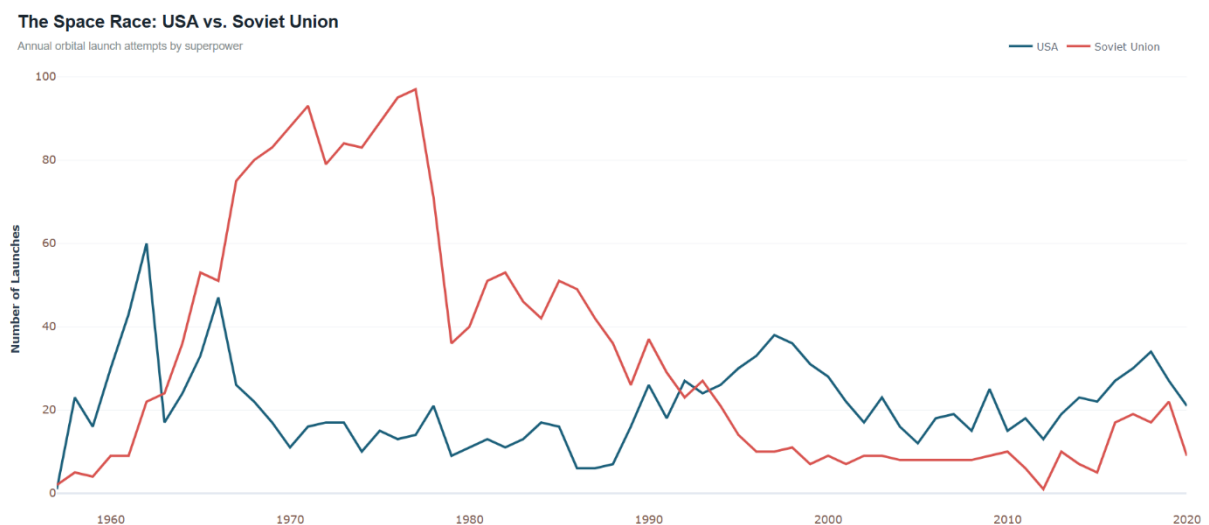
```

        title='', showgrid=False, tickfont=dict(size=14, color='#7D5A50'),
        tickmode='linear', dtick=10
    ),
    yaxis=dict(
        title='Number of Launches',
        title_font=dict(size=14, color='#2C3E50', weight='bold'),
        tickfont=dict(size=14, color='#7D5A50'),
        showgrid=True, gridcolor='#F0F2F6',
        zeroline=True, zerolinecolor='#E2E8F0', zerolinewidth=2
    ),
    legend=dict(
        title_text='', orientation="h",
        yanchor="bottom", xanchor="right",
        font=dict(size=13, color='#5F6B78'),
        bgcolor="rgba(255, 255, 255, 0)"
    ),
    margin=dict(l=20, r=20, t=100, b=20),
    hoverlabel=dict(bgcolor="white", font_size=13, bordercolor="#E2E8F0"),
    plot_bgcolor='rgba(0,0,0,0)', paper_bgcolor='rgba(0,0,0,0)'
)

l_chart.show()

```

Graph:



Interactive Graph Link: [The Space Race: USA vs. Soviet Union](#)

The graph clearly shows that the Soviet Union dominated launch activities for much of the Cold War, particularly in the late 1960s and 1970s, reflecting the USSR's massive investments in military satellites, reconnaissance systems, and intercontinental ballistic missile technology.

Particularly noteworthy is the sharp decline in Soviet launch activity after the late 1980s, which closely coincides with the economic stagnation of the Soviet system and the eventual dissolution of the USSR in 1991, while the United States, through NASA, military programs, and later commercial partnerships

Another key finding is that while the Soviet Union often surpassed the U.S. in raw launch quantity, the United States increasingly focused on technologically complex and impactful missions such as the Apollo lunar program and Space Shuttle operations, suggesting that launch volume alone does not fully reflect scientific or technological leadership during the Space Race era.

Annual Leadership in Global Space Launch Activity

In this section, we are going to examine which country recorded the most space launches in each year to observe how leadership in space exploration has shifted over the course of history.

This analysis illustrates the geopolitical and technological evolution of the global space industry and highlights how dominance has shifted from the Cold War superpowers to a more competitive and internationally distributed space ecosystem.

Preparing the Dataset

To determine the leading country for each year, we first group the dataset by both year and country and then use aggregation and counting operations to calculate the total number of product launches within each combination.

After creating this annual summary at the country level, we apply the *idxmax()* function to the grouped annual data to identify the country with the highest number of launches for each year, thereby determining the annual “leader” in global launches.

This approach illustrates an important data science technique for ranking entities within grouped time-series datasets, which is frequently used in trend analysis, market leadership studies, and time-series comparisons.

```

cntry_and_org_data = df.copy()

# Count total launches per country per year
total_launches_per_year = cntry_and_org_data
    .groupby(['Year', 'Country'])
    .size().reset_index()
    .rename(columns={0: 'Total_Launches'})

leading_country_per_year = total_launches_per_year
    .loc[total_launches_per_year
        .groupby('Year')['Total_Launches'].idxmax()]

```

Creating the Bar Graph for every Leading County

code:

```

country_colors = {
    'USA': '#1A5F7A',
    'Russian Federation': '#D9534F',
    'Kazakhstan': '#E89694',
    'China': '#F3A712'
}

c_bar = px.bar(
    leading_country_per_year,
    x='Year',
    y='Total_Launches',
    color='Country',
    color_discrete_map=country_colors,
    custom_data=['Country']
)

hover_html = (
    "<b style='font-size:16px; color:#1A252F'>{%customdata[0]}</b><br><br>" +
    "<span style='color:#7F8C8D'>Year:</span> <b style='color:#7D5A50'>{%x}</b><br>" +
    "<span style='color:#7F8C8D'>Total Launches:</span> <b style='color:#7D5A50'>{%y}</b>" +
    "<extra></extra>"
)

c_bar.update_traces( hovertemplate=hover_html, marker_line_width=0)

c_bar.update_layout(
    template='plotly_white',
    title=dict(
        text='<b>Global Leaders in Space Exploration</b><br>'
        '<span style="font-size:14px; color:#7F8C8D">Country with the highest

```



```

volume of orbital launches per year</span>',
    font=dict(size=22, family='Helvetica Neue, Arial, sans-serif',
color='#1A252F'),
    x=0.01,
    xanchor='left'
),
xaxis=dict(
    title='',
    showgrid=False,
    tickfont=dict(size=14, color='#7D5A50'),
    tickmode='linear',
    dtick=10
),
yaxis=dict(
    title='Number of Launches',
    title_font=dict(size=14, color='#2C3E50', weight='bold'),
    tickfont=dict(size=14, color='#7D5A50'),
    showgrid=True, gridcolor='#F0F2F6',
    zeroline=True,
    zerolinecolor='#E2E8F0',
    zerolinewidth=2
),
legend=dict(
    title_text='',
    orientation="h",
    yanchor="bottom", xanchor="right",
    y=1.02, x=0.98,
    font=dict(size=13, color='#5F6B78'),
    bgcolor="rgba(255, 255, 255, 0)" # Transparent background
),
margin=dict(l=20, r=20, t=100, b=20),
hoverlabel=dict(bgcolor="white", font_size=15, bordercolor="#E2E8F0"),
plot_bgcolor='rgba(0,0,0,0)',
paper_bgcolor='rgba(0,0,0,0)'
)

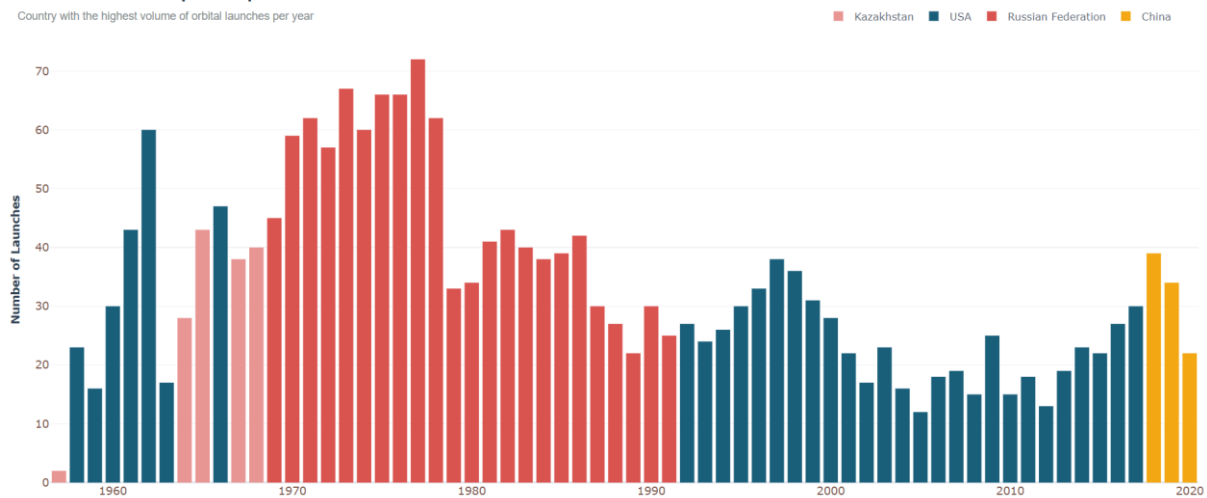
# c_bar.write_html("interactive_graphs/space_race/leading_country_by_year.html")

c_bar.show()

```

Global Leaders in Space Exploration

Country with the highest volume of orbital launches per year



Interactive Graph Link: [Global Leaders in Space Exploration](#)

The graph clearly illustrates how global leadership in space launches has shifted over the course of various geopolitical eras: The United States dominated the early years of the space race, the Soviet Union achieved overwhelming superiority during the decades at the height of the Cold War, and in recent years, China has established itself as the new leading launch nation.

Particularly noteworthy is the long period of Soviet dominance between the late 1960s and the late 1980s, which was driven primarily by the aggressive deployment of military satellites and massive government investment in space technology during the ideological rivalry with the United States.

China's recent rise is particularly significant, as it reflects not only an increasing launch frequency but also the country's broader strategic ambitions to build an independent high-tech space infrastructure through programs such as the BeiDou navigation system, the Tiangong space station, lunar exploration missions, and rapidly expanding commercial satellite networks—suggesting that the modern space race is gradually shifting from Cold War rivalry toward a new era of technological and economic competition centered on Asia.

Conclusion

This report has transformed a comprehensive historical dataset on global space launches into a detailed analytical study of technological advancements, geopolitical competition, and economic development in the industry. By cleaning, grouping, aggregating, and visualizing launch data from 1957 to 2020, we were able to identify key historical trends, such as the Soviet Union's dominance during the Cold War, the market of the U.S. space sector, China's rapid rise in recent years, and the long-term improvement in launch reliability and cost efficiency across the industry.

From a data science perspective, the project demonstrated the practical application of several key statistical methods used in exploration data analysis. Throughout the report, we applied grouping and aggregation operations, data frame merging, filtering, moving averages, trend line analysis, percentile calculations, standard deviation measurements, and time series analysis to uncover hidden relationships within the dataset. Interactive visualizations enabled multidimensional exploration through custom metadata and comparison charts, making complex temporal patterns easier to interpret. Furthermore, statistical analysis played a crucial role in distinguishing between meaningful long-term trends and normal fluctuations. This underscored the fact that data science is not just about creating visualizations, but also about gaining reliable insights, assessing uncertainties, and transforming raw numerical data into understandable historical and technological narratives.

References:

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